

- Q1** Given below are two statements: one is labeled as Assertion A and the other is labeled as Reason R.

Assertion A: Validity checks the real need of system users

Reason R: Completeness checks system user-defined requirements.

In light of the above statements, choose the correct answer from the options given below.

- (A) Both A and R are true and R is the correct explanation of A
 (B) Both A and R are true but R is NOT the correct explanation of A
 (C) A is true but R is false
 (D) A is false but R is true
- Q2** Consider the following statements about the heap sort algorithm:
- A. The MAX-HEAPIFY procedure which runs in $O(\lg n)$ time, is the key to maintaining the max heap property
 B. The BUILD-MAX-HEAP procedure, which runs in $O(\lg n)$ time, produces max-heap from an unordered input array
 C. The MAX-HEAP-INSERT, which runs in $O(\lg n)$ time, implements the insertion operation
 D. The HEAP-INCREASE-KEY procedure runs in $O(n \lg n)$ time, to set the key of a new node to its correct value
- Choose the correct answer from the options given below:
- (A) A, B only (B) A, C only
 (C) B, D only (D) A, B, C, D
- Q3** Consider universe positive integer $X = \{1 \leq n \leq 8\}$, proposition $P = "n \text{ is an even integer}"$, $Q = "(3 \leq n \leq 7) \wedge (n \neq 6)"$. The truth set of $P \leftrightarrow Q$ is
- (A) $\{1, 4\}$ (B) $\{2, 6\}$
 (C) $\{3, 4, 5\}$ (D) $\{1\}$
- Q4** Let $(\mathbb{Z}, +)$ denote the group of all integers under addition. Then the number of all

automorphisms of $(\mathbb{Z}, +)$ is

- (A) 1 (B) 2
 (C) 3 (D) 4

- Q5** Match List I with List II

LIST I	LIST II
A. PERT	I. Used for non repetitive jobs
B. Optimistic time	II. Used for repetitive jobs
C. CPM	III. The shortest possible time to complete the activity if all goes well
D. Pessimistic time	IV. The longest time that an activity takes

Choose the correct answer from the options given below:

- (A) A-III B-IV C-II D-I
 (B) A-IV B-I C-II D-III
 (C) A-I B-III C-II D-II
 (D) A-I B-III C-II D-IV
- Q6** Which of the following is used to determine the cost performance index?
- (A) Budgeted cost of work performed-budget at completion
 (B) Budgeted cost of work performed $\div$ budget at completion
 (C) Budgeted cost of work performed $\div$ Actual cost of work performed
 (D) Budgeted cost of work performed-Actual cost of work performed
- Q7** _____ is intended to show that a system both conforms to its specifications and meets the expectations of the system customer.
- (A) Software specification
 (B) Software design
 (C) Software evaluation
 (D) Software validation

- Q8** A TCP server application is programmed to listen on port P on host S. A TCP client is connected to the TCP server over the network, Consider that while TCP connection is active the server is crashed and rebooted. Assume that the client does not use TCP keepalive timer. Which of the following behaviour/s is/are possible?
- Statement I: The TCP application server on S can listen on P after reboot.



Statement II: If client sends a packet after the server reboot, it will receive the RST segment. In the light of the above statements, choose the correct answer from the options given below.

- (A) Both Statement I and Statement II are true
(B) Both Statement I and Statement II are false
(C) Statement I is true but Statement II is false
(D) Statement I is false but Statement II is true

Q9 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: I/O protection is ensured by a hardware trap

Reason R: I/O interrupt caused by the condition like I/O completion and device malfunction occurring within the I/O devices

In the light of the above statements, choose the correct answer from the options given below.

- (A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false
(D) A is false but R is true

Q10 Match List I with List II

LIST I		LIST II	
A.	Parallel FFT	I.	$O(n^2)$
B.	Iterative FFT	II.	$O(n)$
C.	Evaluation of polynomial at n points by Horner method	III.	$O(\lg n)$
D.	Product of two polynomials that are represented in point value form	IV.	$O(n \lg n)$

Choose the correct answer from the options given below:

- (A) A-III B-I C-II D-III
(B) A-II B-I C-III D-IV
(C) A-III B-IV C-I D-II
(D) A-II B-III C-IV D-I

Q11 Match List I with List II

LIST I		LIST II	
A.	Bezier curves	I.	3 dim objects that have translation rotational symmetry
B.	B-Splines	II.	Fractal geometry
C.	Sweep representations	III.	Bias and tension
D.	Natural objects	IV.	painting drawing CAD

Choose the correct answer from the options given below:

- (A) A-IV B-I C-III D-II

(B) A-II B-III C-IV D-I

(C) A-IV B-III C-I D-II

(D) A-II B-IV C-I D-III

Q12 Given below are two statements:

Let $f(n)$ and $g(n)$ be asymptotically positive functions. The following conjectures are given

Statement I:

$$f(n) \geq 1 \text{ and } f(n) = O(g(n)) \Rightarrow g(n) = \Omega(f(n))$$

Statement II:

$$f(n) = O(g(n)) \Rightarrow \lg(f(n)) = O(\lg(g(n)))$$

where $\lg(g(n)) \geq 1$ for all sufficient large n .

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both Statement I and Statement II are correct
(B) Both Statement I and Statement II are incorrect
(C) Statement I is correct but Statement II is incorrect
(D) Statement I is incorrect but Statement II is correct

Q13 Consider the following statements:

S1: LRU page replacement algorithm suffers from the belady's anomaly

S2: Shortest remaining time first scheduling may cause starvations

S3: Stack is shared by all threads in a process

- (A) S1 S2 and S3 are true
(B) S1, S3 false and S2 is true
(C) S1, S2 are false and S3 is true
(D) S1,S2 and S3 are false

Q14 The total cost of retrieving records in sorted order using an unclustered B^+ tree is

(P- Average number of records per data page

N - Data pages

F- Ratio of the size of a data entry to the size of a data record)

- (A) $(F * N) + P$ (B) $(F + P) * N$
(C) $F * N * P$ (D) $F + P / N$

Q15 Match List I with List II



LIST I		LIST II	
A.	Scenario testing	I.	To verify the I/O behavior of text object
B.	Regression testing	II.	user acceptance methodology
C.	Component testing	III.	No new bugs after changes in program
D.	Beta testing	IV.	The documentation of a use case

Choose the correct answer from the options given below:

- (A) A-IV B-III C-II D-I
 (B) A-II B-I C-III D-IV
 (C) A-IV B-III C-I D-II
 (D) A-III B-I C-IV D-II

Q16 A. If some NP-complete problem P is in $\mathbb{P}$ that

$\mathbb{P} = \text{NP}$

B. TSP is in NP

C. SAT is in NP

D. Hamilton circuit problem is not NP-complete

Choose the correct answer from the options given below:

- (A) A, B and C only
 (B) B, C and D only
 (C) C, D and A only
 (D) D, A and B only

Q17 A B-tree used as an index for a large database table has four levels including the root node. If a new key is inserted in this index, then the maximum number of nodes that could be newly created in the process is

- (A) 5 (B) 4
 (C) 3 (D) 2

Q18 Suppose a circular queue of capacity $n-1$ elements is implemented with an array of n elements. Assume that the insertion and deletion operations are carried out using REAR and FRONT as array index variable respectively. Initially, $\text{REAR} = \text{FRONT} = 0$. The conditions to detect queue empty and queue full are

(A) EMPTY: $\text{REAR} == \text{FRONT}$

FULL: $(\text{REAR} + 1) \bmod n == \text{FRONT}$

(B) EMPTY: $(\text{FRONT} + 1) \bmod n == \text{REAR}$

FULL: $(\text{REAR} + 1) \bmod n == \text{FRONT}$

(C) EMPTY: $(\text{REAR} + 1) \bmod n == \text{FRONT}$

FULL: $\text{REAR} == \text{FRONT}$

(D) EMPTY: $\text{REAR} == \text{FRONT}$

FULL: $(\text{FRONT} + 1) \bmod n == \text{REAR}$

Q19 Consider the following statements about the software product line system:

Statement I: At the interaction level, components provide an operator display interface and an interface with the communication system used.

Statement II: At the I/O management level, components handle operator authentication, report generator and query manager.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both Statement I and Statement II are correct
 (B) Both Statement I and Statement II are incorrect
 (C) Statement I is correct but Statement II is incorrect
 (D) Statement I is incorrect but Statement II is correct

Q20 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: A process involves a library function to create a thread.

Reason R: The threads make system calls to convey their resource and I/O requirement to the Kernel.

In the light of the above statements, choose the correct answer from the options given below.

- (A) Both A and R are true and R is the correct explanation of A
 (B) Both A and R are true but R is NOT the correct explanation of A
 (C) A is true but R is false
 (D) A is false but R is true

Q21 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: The AVL trees are more balanced as compared to Red Black trees, but they may cause more rotations during insertion and deletion

Reason R: A Red Black tree with n nodes has height that is greater than $2 \log_2(n + 1)$ and



the AVL tree with n nodes has height less than $\log_{\Phi}(\sqrt{5}(n+2)) - 2$ (where Φ is golden ratio)

In the light of the above statements, choose the correct answer from the options given below.

- (A) Both A and R are correct and R is the correct explanation of A
 (B) Both A and R are correct and R is NOT the correct explanation of A
 (C) A is true but R is false
 (D) A is false but R is true

Q22 Match List I with List II

List I	List II
A. Physical layer	I. Transforming the raw bits in the form of a frame for transmission
B. Data Link Layer	II. Control and monitoring of subnet
C. Network Layer	III. Transmission of raw bits over communication channel
D. Transport Layer	IV. Datagrams transmission data through connection-oriented or connectionless using datagram

Choose the correct answer from the options given below:

- (A) A-III B-II C-I D-IV
 (B) A-II B-III C-I D-IV
 (C) A-III B-I C-II D-IV
 (D) A-II B-IV C-I D-III

Q23 Let $R = \{x, x \in N, x \text{ is multiple of 3 and } x \leq 100\}$ and $S = \{x, x \in N, x \text{ is a multiple of 5 and } x < 100\}$. What is the number of elements in $(R \cap S) \times (S \cap R)$?

- (A) 36 (B) 33
 (C) 20 (D) 6

Q24 let $R(A, B, C, D)$ be a relational schema with following function dependencies:

$A \rightarrow B, B \rightarrow C$

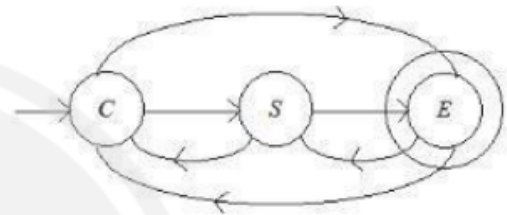
$C \rightarrow D$ and $D \rightarrow B$

The decomposition of R into

$(A, B)(B, C)(B, D)$

- (A) gives a lossless join, and is dependency preserving
 (B) gives lossless join, but is not dependency preserving
 (C) does not give a lossless join, but is dependency preserving
 (D) does not give a lossless join and is not dependency preserving

Q25 Consider the following finite automata F1 that accepts a language L



Let F2 be a finite automata which is obtained by reversal of F1. Then which of the following is correct?

- (A) $L(F1) \neq L(F2)$
 (B) $L(F1) = L(F2)$
 (C) $L(F1) \leq L(F2)$
 (D) $L(F1) \geq L(F2)$

Q26 Given below are two statements:

Which of the following statement/s is/are correct with respect to virtual memory

Statement I: Address translation is performed for every logical address used during the execution of a program

Statement II: A program can execute only when all of its components are loaded in the memory
 In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both Statement I and Statement II are correct
 (B) Both Statement I and Statement II are incorrect
 (C) Statement I is correct but Statement II is incorrect
 (D)



Statement I is incorrect but Statement II is correct

- Q27** There are M points on one straight line AB and n points on another straight line AC none of them being A . How many triangles can be formed with these points as vertices?

(A) $mn(m+n-2)$ (B) $\frac{1}{2}mn(m+n-2)$
(C) $\frac{1}{2}mn(m+n-1)$ (D) $mn(m+n-1)$

- Q28** Given below are two statements:

Statement I: Fuzzifier is a part of a fuzzy system

Statement II: Inference engine is a part of fuzzy system

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both Statement I and Statement II are correct
(B) Both Statement I and Statement II are incorrect
(C) Statement I is correct but Statement II is incorrect
(D) Statement I is incorrect but Statement II is correct

- Q29** Consider the following statements:

A. A database design is in BCNF if each member of the set of relation schemas that constitutes the design is in BCNF

B. A BCNF schema can have transitive dependency

C. It is always possible to obtain a 3 NF design without sacrificing a lossless join

D. There are multivalued dependencies in 4 NF

- (A) A, B and C only
(B) B, C and D only
(C) A, B and D only
(D) A, C and D only

- Q30** Which of the following is not a property of a good system for representation of knowledge in a particular domain?

- (A) Presentation adequacy
(B) inferential adequacy
(C) Inferential efficiency
(D) acquisitional efficiency

- Q31** What will be the output of the following code?

```
# include < stdio. h>
int main () {
int a, b, c;
a=o×10; b=o10;
c=a+b;
printf("%d", c);
return 0;
}
```

- (A) 20 (B) 24
(C) Garbage (D) error

- Q32** What is the 4+4 fit binary fixed point equivalent of $-(3.72)_{10}$?

- (A) 0011.1100 (B) 0011.1010
(C) 1100.0100 (D) 0011.1011

- Q33** Let N denote the set of all natural numbers and R be the relation on $N \times N$ defined by $(a, b)R(c, d)$, if $ad(b+c) = bc(a+d)$. Then R is

- (A) Symmetric only
(B) Reflexive only
(C) Transitive only
(D) An equivalence relation

- Q34** The maximum yield length of the following CNF CFG is

$S \rightarrow AB$
 $A \rightarrow CD$
 $B \rightarrow e$
 $C \rightarrow a$
 $D \rightarrow b$

- (A) 8 (B) 7
(C) 4 (D) 5

- Q35** Let $\oplus$ denote XOR operation. Let 1 and 0 denote the binary constants and F is the Boolean expression over two variables P and Q ($F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q))$

$\oplus ((P \oplus Q) \oplus (Q \oplus 0))$

Which of the following is equivalent expression to F ?

- (A) $P \oplus Q$
(B) $P + Q$
(C) $\overline{P + Q}$
(D) $\overline{P \oplus Q}$



- Q36** Which of the following scenario may lead to an irrecoverable error in a database system?
- (A) A transaction writes a data item after it is read by an uncommitted transaction
- (B) A transaction reads a data item after it is read by an uncommitted transaction
- (C) A transaction reads a data item after it is written by a committed transactions
- (D) A transaction reads a data item after it is written by an uncommitted transaction.

- Q37** An observational technique that can be used to understand operational process and help to derive requirement for software to support operational process is known as
- (A) Requirement specification
- (B) Structural specification
- (C) Ethnography
- (D) Natural language specification

- Q38** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

A virtual memory system uses first-in first-out page replacement policy and allocates a fixed number of frames to a process

Assertion A: Increasing number of page frames allocated to a process sometimes increases the page fault rate.

Reason R: Some programs do not exhibit locality of reference.

In the light of the above statements, choose the correct answer from the options given below.

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is NOT the correct explanation of A
- (C) A is true but R is false
- (D) A is false but R is true

- Q39** Match List I with List II

List-I	List-II
A. CDMA	I. It provides mobile internet connection with faster data transfer rates

B. GSM	II. It allows users to connect to a network or to other devices over wireless channel
C. UMTS	III. Accessing mechanism for multiple transmitters over a single channel
D. WiFi	IV. It is cellular technology employs hybrid of FDMA and TDMA

Choose the correct answer from the options given below:

- (A) A-III B-IV C-II D-I
- (B) A-III B-IV C-I D-II
- (C) A-II B-III C-IV D-I
- (D) A-II B-I C-IV D-III

- Q40** Consider the following language :

L

$= \{w$

$\in \{a, b, c\}^* : n_a(w) + n_b(w) = n_c(w)\}$

L is

- (A) Context free but not linear
- (B) Not context free
- (C) Context free and linear
- (D) Linear

- Q41** Which of the following is not a mutation operator in a genetic algorithm?

- (A) Random resetting
- (B) Scramble
- (C) Inversion
- (D) Difference

- Q42** Consider the following table of arrival time and burst time for three processes P0, P1 P2:

Process	arrival time	Burst time
P0	0 ms	7
P1	1 ms	3
P2	2 ms	7

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of a process. What is the average waiting time for the three processes?



- (A) 3 ms (B) 3.67 ms
(C) 4.47 ms (D) 4 ms

Q43 Which is not the component of the natural language understanding process?

- (A) Morphological analysis
(B) Semantic analysis
(C) Pragmatic analysis
(D) Meaning analysis

Q44 Consider a hash table of size seven with starting index zero and a hash function $(6x + 3) \bmod 4$. Assuming the hash table is initially empty. Which of the following is the content of the table when the sequence 1, 3, 8, 10, 5, is inserted into the table using closed hashing? Here " _ " denotes an empty location in the table.

- (A) 1,3,8,10,5,_ (B) 3,8,1,_,10,5
(C) _,3,8,1,_,10,5 (D) _,1,3,8,10,5_

Q45 A three dimensional array in C++ is declared as `int A[a][b][c]`. Consider that array elements are stored in row major order and indexing begin from 0. Here the address of an item at the location `A[r][s][t]` computed in terms of word length w of an integer is

- (A) $\&A[0][0][0] + w(b \cdot c \cdot s + c \cdot r + t)$
(B) $\&(A[0][0][0] + w(b \cdot c \cdot r^* + c \cdot s + t))$
(C) $\&A[0][0][0] + w(a \cdot b \cdot r^* \cdot r^* + c \cdot s + t)$
(D) $\&A[0][0][0] + w(a \cdot b \cdot s + c \cdot r + t)$

Q46 Match List I with List II

LIST I		LIST II	
A.	RAID level 2	I.	Bit interleaved parity
B.	RAID level 4	II.	Block interleaved distributed parity
C.	RAID level 5	III.	Error correcting parity
D.	RAID level 3	IV.	Block interleaved parity

Choose the correct answer from the options given below:

- (A) A-IV B-III C-I D-II
(B) A-III B-IV C-II D-I
(C) A-III B-I C-II D-IV
(D) A-I B-III C-IV D-I

Q47

Match List I with List II

LIST I		LIST II	
A.	Expert system	I.	Decision tree
B.	Fuzzy system	II.	Scramble
C.	Operator in genetic algorithm	III.	Inference engine
D.	Supervised technique	IV.	Mycin

Choose the correct answer from the options given below:

- (A) A-IV B-I C-III D-II
(B) A-III B-IV C-II D-I
(C) A-IV B-III C-II D-I
(D) A-I B-II C-III D-IV

Q48 Consider following statements:

- A. A context free language is generated by LR(0) grammar if and only if it is accepted by a deterministic pushdown automata and has prefix property
B. If M_1 is the single tape TM simulating multilape TMM , then time taken by M_1 to simulate n moves is (n^3)
C. Push down automata behaves like a Turing machine when it has one auxiliary memory.
D. $L = \{a^n b^n c^n : n \geq 1\}$ is not context free but context sensitive.

Choose the correct answer from the options given below:

- (A) A, B and C only
(B) A, B only
(C) D only
(D) B, C only

Q49 Consider a popular sports news site. At a given moment, 20,000 concurrent users submit a request (a transaction, T) once every 2 minutes on average. Each transaction requires the webapp to download a new article that on average has 3 k bytes in length. What is the throughput?

- (A) 8 megabits per second
(B) 4 megabits per second
(C) 6 megabits per second
(D) 2 megabits per second

Q50 How many integral solutions are there to $x + y + z + w = 29$, where $x \geq 1, y \geq 2, z \geq 3$ and $w \geq 0$?



- (A) 2400 (B) 2600
(C) 2800 (D) 3000

Q51 Let $R(A, B, C, D, F)$ be a relational schema with following functional dependencies:
 $C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B$. Which of the following is a key for R?

- (A) CD (B) EC
(C) AE (D) AC

Q52 What is the output of following code?

```
main ()
struct s1
{ char*z;
int i;
struct s1 * p;
}
static struct s1 a [ ] = {
{"Nagpur", 1, a+1}
{"Raipur", 2, a+2}
{"Kanpur", 3, a}
};
struct s1* ptr =a;
printf("%s %s %s \n", a[0].z, ptr ->z, a[2].p ->z);
}
```

- (A) Nagpur Raipur Kanpur
(B) Nagpur Nagpur Nagpur
(C) Kanpur Kanpur Kanpur
(D) Error

Q53 Which one of the following is NOT a part of ACID properties of a database transaction?

- (A) Atomicity
(B) Consistency
(C) Isolation
(D) Deadlock-freedom

Q54 Match List I with List II

LIST I			LIST II	
A.	Critical region	I.	Hoare's monitor	
B.	Wait/signal	II.	Mutual exclusion	
C.	Working set	III.	Principle of locality	
D.	Deadlock	IV.	Circular wait	

Choose the correct answer from the options given below:

- (A) A-IV B-I C-III D-II

- (B) A-II B-I C-III D-IV
(C) A-I B-II C-III D-IV
(D) A-IV B-III C-I D-II

Q55 Consider two hosts P and Q that are connected through a router R. The maximum transfer unit (MTU) value of the link between P and R is 1500 bytes and between R and Q is 820 bytes. A TCP segment of size 1400 bytes is transferred from P to Q through R with IP identification value of 0×1234 . Assume that IP header size is 20 bytes. Further the packet is allowed to be fragmented that is Don't fragment (DF) flag in the IP Header is not set by P. Which of the following statement/s is/are true?

- A. Two fragments are created at R and IP datagram size carrying the second fragment is 620 bytes
B. If the second fragment is lost, then R resends the fragment with IP identification value of 0×1234
C. If the second fragment lost, then P requires to resend the entire TCP segment.
D. TCP destination port can be determined by analyzing the second fragment only.

Choose the correct answer from the options given below:

- (A) A, B and C only
(B) A and C only
(C) C and D only
(D) B and D only

Q56 Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A: Dendral is an expert system

Reason R: The rationality of an agent is not related to its reaction to the environment.

In the light of the above statements, choose the correct answer from the options given below.

- (A) Both A and R are true and R is the correct explanation of A
(B) Both A and R are true but R is NOT the correct explanation of A
(C) A is true but R is false



(D) A is false but R is true

Q57 In a multiuser operating system, 20 requests are made to use a particular resource per hour, on an average. The probability that no request is made in 45 minutes is

- (A) e^{-15} (B) e^{-5}
(C) $1 - e^{-5}$ (D) $1 - e^{-10}$

Q58 At a particular time of computation, the value of a counting semaphore is 7. Then 20 P operations and 'x' V operations were completed on this semaphore. If the final value of semaphore is 5. x will be

- (A) 15 (B) 22
(C) 18 (D) 13

Q59 Which of the following is not a solution representation in a genetic algorithm?

- (A) Binary valued (B) Real valued
(C) Permutation (D) Combinations

Q60 Which phase of compiler checks the grammar of programming?

- (A) Code optimization
(B) Semantic analysis
(C) Code generators
(D) Syntax analysis

Q61 Given the basic E R diagram and relational model, which of the the following is incorrect?

- (A) An attribute of an entity can have more than one value
(B) An attribute of an entity can be composite
(C) In a row of relational table, an attribute can have more than one value
(D) In a row of a relational table, an attribute can have exactly one value or a NULL value

Q62 A processor chip is used for application in which 30 % of execution time is spent on floating point addition. For the new model of the processor, the design team has come up with redesign the floating point adder to make it twice as fast. What will be possible maximum speed up by this redesign?

- (A) 2.0 (B) 1.06

(C) 1.18

(D) 2.5

Q63 Given below are two statements:

Which of the following concurrency control protocol ensures both conflict serializability and freedom from deadlock?

Statement I: Two phase locking

Statement II: Timestamp ordering

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both Statement I and Statement II are correct
(B) Both Statement I and Statement II are incorrect
(C) Statement I is correct but Statement II is incorrect
(D) Statement I is incorrect but Statement II is correct

Q64 Consider the following statements

- A. C-Fuzzy means clustering is a supervised method of learning
B. PCA is used for dimension reduction
C. Apriori is not a supervised technique
D. When a machine learning model becomes so specially tuned to its exact input data that it fails to generalize to other similar data it is called underfitting

Choose the correct answer from the options given below:

- (A) A and B (B) B and C
(C) C and D (D) D and A

Q65 Sutherland Hodgeman method is used on

- (A) Smooth curves
(B) Line segment
(C) Convex polygons
(D) Concave polygons

- Q66** A. The set of turning machine codes for TM's that accept all inputs that are palindromes (possible along with some other inputs) is decidable
B. The language of codes for TM's M that when started with blank tape, eventually write a 1 somewhere on the tape is undecidable
C. The language accepted by a TMM is LM is



always recursive

D. Post's correspondence problem is undecidable

Choose the correct answer from the options given below:

(A) A, B and C only

(B) B, C and D only

(C) A and C only

(D) B and D only

Q67 Let $G = \frac{R}{\{0\}}$ and $H = \{-1, 1\}$ be groups

under the multiplication. Then, the map

$\phi : G \rightarrow H$ defined by $\phi(x) = \frac{x}{|x|}$ is

(A) Not a homomorphism

(B) A one-one homomorphism, which is not onto

(C) An onto homomorphism, which is not one to one

(D) An homomorphism

Q68 Which of the following parser is most powerful parser?

(A) Operator precedence

(B) SLR

(C) Canonical LR

(D) LALR

Q69 What is x in the following program?

```
#include <stdio.h>
int main ()
{typedef (* (* arrfptr [3])) [10];
arrfptr x;
return 0;
}
```

(A) x is a pointer

(B) x is a array of three pointer

(C) x is an array of three function pointer

(D) Error in x declaration

Q70 Find the sum of all four digit numbers formed using the digits 1,2,4 and 6 .

(A) 86,658

(B) 88,8858

(C) 91,958

(D) 93,358

Q71 Consider the following statements:

A. Dynamic metrics are collected by measurements made of a program in execution

B. Static metrics are collected by measurements made of representations of the system

C. The assessment of software quality is an objective process

D. An important part of quality assurance in the selection of standards that should apply to the software development process.

Choose the correct answer from the options given below:

(A) A, B, C only

(B) B, C, D only

(C) A, C, D only

(D) A, B, D only

Q72 The following table shows the time between failures for a software :

Error number	1	2	3	4	5
time since last failure (Hours)	6	4	8	5	6

The reliability of the system for one hour operation assuming an exponential model is-

(A) $e^{-9/29}$

(B) $e^{-7/29}$

(C) $e^{-5/29}$

(D) $e^{-3/29}$

Q73 Which of the following statement is correct?

(A) Ackermann's function is primitive recursive.

(B) $L = \{a^n b^k c^{n+k} : n \geq 0, k \geq 0\}$ is regular language.

(C) $L = \{a^n b^J : n = J^2\}$ is not context free language

(D) For every context sensitive language L not including λ , there exists some linear bounded automata M such that $L \neq L(M)$.

Q74 How will you free the memory allocated by the following program?

```
#include <stdio.h>
#include <stdlib.h>
#define MAXROW 3
#define MAXCOL 4
int main ()
{
int ** p, i, j;
p= (int **) malloc (MAXROW * size of (int*));
return 0;
}
```

(A) memfree (int p);

(B) dealloc (p);



- (C) malloc (p,0);
(D) free (p);

- Q75** Which is not a basic approach to the problem of conflict resolution in a production system?
(A) Assigning a preference based on the rule that matched
(B) Assigning a preference based the object that matched
(C) Assigning a preference based on the action that the matched rule would perform
(D) Assigning a preference based on the action that the matched object would perform

- Q76** Match List I with List II

LIST I		LIST II	
A.	$A \triangle B$	I.	$(A - B) \cup (A - C)$
B.	$A - (B \cup C)$	II.	$(A - B) \cap (A - C)$
C.	$A - (B \cap C)$	III.	$(A - B) \cup (B - A)$
D.	$A \cap (B - C)$	IV.	$(A \cap B) - (A \cap C)$

Choose the correct answer from the options given below

- (A) A-III B-II C-I D-IV
(B) A-II B-III C-IV D-I
(C) A-IV B-III C-I D-II
(D) A-IV B-I C-III D-II
- Q77** The clipping process in computer graphics is used for
(A) Adding graphics
(B) Copying
(C) Zooming
(D) Removing objects and lines
- Q78** Consider a disk system with cylinders. The request to access the cylinders occurs in the following sequence:
4, 34, 10, 7, 19, 73, 2, 15, 6, 20
Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from one cylinder to adjacent one and shortest seek time first policy is used?
(A) 119 ms (B) 120 ms
(C) 142 ms (D) 146 ms

- Q79** Consider the rectangle with vertices (0, 0), (0, 2), (3, 0), (3, 2). There is scaling of 2 towards x-axis and 3 towards y-axis. The new coordinates of the rectangle are
(A) (0, 0), (6, 0), (0, 4), (6, 4)
(B) (0, 0), (6, 0), (0, 4), (3, 2)
(C) (0, 0), (6, 0), (0, 6), (6, 6)
(D) (0, 0), (4, 0), (0, 6), (4, 6)

- Q80** Which of the following transforms in 2 dimension is used to resize a 2-dimensional object?
(A) Translation (B) Rotation
(C) Scaling (D) Shearing

- Q81** In the standard Ethernet with transmission rate of 10 Mbps, assume that the length of the medium is 2500 m and size of a frame is 512 bytes. The propagation speed of a signal in a cable is normally 2×10^8 m/s. The transmission delay and propagation delay are
(A) 25. $25\mu s$ and 51. $2\mu s$
(B) 51. $2\mu s$ and 12. $5\mu s$
(C) 10. $24\mu s$ and 50. $12\mu s$
(D) 12. $5\mu s$ and 51. $2\mu s$

- Q82** Consider the following statements.
A. The identity is unique in any monoid.
B. A monoid is a group if there exists inverse of each element of monoid.
C. Semi group has closure, associative and identity properties.
D. Quasi group has closure property.
(A) A, B and D only
(B) B, C and D only
(C) A, B and C only
(D) A, C and D only

- Q83** What is the output of following code?
main ()
{static float a[] = {13.24, 1.5, 4.5, 5.4, 3.5}
float *j, *k;
j=a;
k = a+4
j =j*2;
k =k/2



```
printf ("%f %f", *j, *k);
}
```

- (A) 13. 25, 4. 5
 (B) 1. 5, 3. 5
 (C) 13. 24, 1. 5, 4. 5, 5. 4, 3. 5
 (D) Illegal use of pointer in main function

- Q84** If $A = \{4n + 2 \mid n \text{ is a natural number}\}$ and $B = \{3n \mid n \text{ is a natural number}\}$. Which of the following is correct for $A \cap B$?
 (A) $\{12n^2 + 6n \mid n \text{ is a natural number}\}$
 (B) $\{24n - 12 \mid n \text{ is a natural number}\}$
 (C) $\{60n + 30 \mid n \text{ is a natural number}\}$
 (D) $\{12n - 6 \mid n \text{ is a natural number}\}$

- Q85** Match List I with List II

	List I		List II
A.	The running time of a straight forward recursive method to compute nth Fibonacci number F_n	I.	$O(n^2)$
B.	The running time to compute F_n using memoization.	II.	$O(\log n)$
C.	The running time to compute Fibonacci number F_n using only integer addition and multiplication.	III.	$O(n)$
D.	The running time to determine an optimal bitonic tour.	IV.	$\Theta(\phi^n)$

Choose the correct answer from the options given below:

- (A) A-I B-III C-IV D-II
 (B) A-IV B-III C-II D-I
 (C) A-I B-II C-IV D-III
 (D) A-IV B-II C-III D-I
- Q86** Given below are two statements:
 Statement I: If f and g are two functions and $f = O(g)$ but $g \neq o(f)$, we say that the growth rate of g is smaller than that of f

Statement II: The class of all decision problems decided by a TM in exponential time, that is $O(2^k)$, k being a constant.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both Statement I and Statement II are correct
 (B) Both Statement I and Statement II are incorrect
 (C) Statement I is correct but Statement II is incorrect
 (D) Statement I is incorrect but Statement II is correct

- Q87** 256 Mb DRAM is organized as a $32M \times 8$ memory externally and as a $16K \times 16K$ square array internally. Each row must be refreshed at least once every 50 milli second to forestall loss of data; refreshing one row takes 100 nanoseconds. What fraction of the total memory bandwidth is lost to refresh cycles?
 (A) 6.6% (B) 3.3%
 (C) 9.9% (D) 4.3%

- Q88** Given below are two statements:
 Statement I: subsystem models show logical grouping of objects into coherent subsystem
 Statement II: State machine models show how objects change their states in response to events.
 In the light of the above statements, choose the most appropriate answer from the options given below.
 (A) Both Statement I and Statement II are correct
 (B) Both Statement I and Statement II are incorrect
 (C) Statement I is correct but Statement II is incorrect
 (D) Statement I is incorrect but Statement II is correct

- Q89** Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.
 Assertion A: It is possible to create doubly linked list using only one pointer with every node.



Reason R: By storing the XOR of the addresses of the previous and next nodes.

In the light of the above statements, choose the most appropriate answer from the options given below.

- (A) Both A and R are true and R is the correct explanation of A
- (B) Both A and R are true but R is NOT the correct explanation of A
- (C) A is true but R is false
- (D) A is false but R is true

- Q90** The compiler for high level language that runs on one machine and produces code for other machine is called-
- (A) Cross compiler
 - (B) Multipass compiler
 - (C) Optimizing Compiler
 - (D) More than one of the above
 - (E) None of the above

Directions (91-95) Read the following passage and answer the given questions.

Consider the following program fragment that deals with a table T with 17 rows and 1024 columns, computing an average for each column and printing it to screen (i is row index and j is column index):

```
for j= [ 0...1023] {
temp =0;
for i= [0...16]
temp = temp + T[i][j];
print (temp/17.0); }
```

T [i] [j] and temp are 32 bit floating point values and memory is word addressable. The temporary variable temp is kept in a processor register so access to temp does not involve a memory reference. The main memory is page and holds 16 pages of size 1024 words, the page replacement policy is "least recently used ". If T is stored in the virtual address space in row major format.

- Q91** Consider again that T is stored in column-major format, what is the main memory hit ratio?
- (A) 80%
 - (B) 95.6%

- (C) 97.8%
- (D) 99.9%

- Q92** Consider that T is stored in column major format, how many page faults will be encountered?
- (A) 14
 - (B) 15
 - (C) 16
 - (D) 17
- Q93** How many page faults will be encountered?
- (A) 16,402
 - (B) 17,408
 - (C) 18,208
 - (D) 18,608
- Q94** What is fault ratio of row major to column major arrangements ?
- (A) 1024: 1
 - (B) 1301: 1
 - (C) 1240: 1
 - (D) 9107: 8
- Q95** What is the main memory hit ratio?
- (A) 0
 - (B) 1
 - (C) 2
 - (D) 3

Directions (96-100) Read the following passage and answer the given questions.

IP datagram has arrived with following partial information in the header (in hexadecimal)
45000054000300002006....

- Q96** How many more routers can the packet travel to?
- (A) 22
 - (B) 26
 - (C) 30
 - (D) 32
- Q97** What is the protocol of the payload being carried by the packet?
- (A) ICMP
 - (B) SCTP
 - (C) TCP protocol
 - (D) IGMP
- Q98** What is the size of datagram?
- (A) 64 bytes
 - (B) 74 bytes
 - (C) 84 bytes
 - (D) 104 bytes
- Q99** What is the header size?
- (A) 10 bytes
 - (B) 20 bytes
 - (C) 30 bytes
 - (D) 40 bytes
- Q100** What is the efficiency of this datagram?
- (A) 76.19%
 - (B) 80.50%
 - (C) 82.24%
 - (D) 85.45%



Answer Key

Q1 (B)
Q2 (B)
Q3 (A)
Q4 (B)
Q5 (D)
Q6 (C)
Q7 (D)
Q8 (A)
Q9 (D)
Q10 (C)
Q11 (C)
Q12 (A)
Q13 (B)
Q14 (B)
Q15 (C)
Q16 (A)
Q17 (A)
Q18 (A)
Q19 (A)
Q20 (B)
Q21 (C)
Q22 (C)
Q23 (A)
Q24 (A)
Q25 (B)
Q26 (C)
Q27 (B)
Q28 (A)
Q29 (D)
Q30 (A)
Q31 (B)

Q32 (C)
Q33 (D)
Q34 (C)
Q35 (D)
Q36 (D)
Q37 (C)
Q38 (B)
Q39 (B)
Q40 (A)
Q41 (D)
Q42 (B)
Q43 (D)
Q44 (D)
Q45 (B)
Q46 (B)
Q47 (C)
Q48 (C)
Q49 (B)
Q50 (B)
Q51 (B)
Q52 (B)
Q53 (D)
Q54 (B)
Q55 (B)
Q56 (C)
Q57 (A)
Q58 (C)
Q59 (D)
Q60 (D)
Q61 (C)
Q62 (C)



Q63 (D)
Q64 (B)
Q65 (C)
Q66 (D)
Q67 (C)
Q68 (C)
Q69 (C)
Q70 (A)
Q71 (D)
Q72 (C)
Q73 (C)
Q74 (D)
Q75 (D)
Q76 (A)
Q77 (D)
Q78 (A)
Q79 (C)
Q80 (C)
Q81 (B)

Q82 (A)
Q83 (D)
Q84 (D)
Q85 (B)
Q86 (B)
Q87 (B)
Q88 (A)
Q89 (A)
Q90 (A)
Q91 (D)
Q92 (D)
Q93 (B)
Q94 (A)
Q95 (A)
Q96 (D)
Q97 (C)
Q98 (C)
Q99 (B)
Q100 (A)



Hints & Solutions

Q1 Text Solution:

Both A and R are true but R is NOT the correct explanation of A, because R does not directly explain A. Completeness checks requirements, while validity checks whether the system addresses the real needs of the users, which is broader than just checking requirements.

Q2 Text Solution:

The correct answer is **Option B (A, C only)**.

- **A: Correct.** The **MAX-HEAPIFY** procedure ensures that the max-heap property is maintained by comparing a node with its children and making necessary swaps. It runs in $O(\log n)$ time because it might need to traverse from the node down to the leaf level in the worst case.
- **B: Incorrect.** The **BUILD-MAX-HEAP** procedure transforms an unordered array into a max-heap. While MAX-HEAPIFY runs in $O(\log n)$ time, the BUILD-MAX-HEAP process works on multiple nodes and, overall, runs **$O(n)$ time**, not $O(\log n)$ time.
- **C: Correct.** The **MAX-HEAP-INSERT** operation inserts a new element into the heap and then "bubbles up" the element to restore the heap property. This process takes $O(\log n)$ time because it may need to move the element up the tree, which has a height of $\log n$.
- **D: Incorrect.** The **HEAP-INCREASE-KEY** procedure adjusts the value of a node and restores the max-heap property by moving the node upward. This operation only needs $O(\log n)$ time since it only traverses up the tree, not $O(n \log n)$.

Q3 Text Solution:

The correct answer is **Option A ({1,4})**.

The truth set of $P \leftrightarrow Q$,

$P \leftrightarrow Q$ includes integers for which both propositions are either true or false.

For $n=1$, Both P and Q are false (true for $P \leftrightarrow Q$).

For $n=4$, Both P and Q are true (true for $P \leftrightarrow Q$).

And for all other values in given set one of them (either P or Q) is true which makes the result of $P \leftrightarrow Q$ is false.

Thus, the truth set is $\{1, 4\}$.

Q4 Text Solution:

The correct answer is **Option B (2)**.

The group $(Z, +)$ has automorphisms defined by $f(n) = kn$, where k must be a unit (invertible element) in Z .

In the group of all integers under addition, there are only two automorphisms. One is the identity map (each number maps to itself i.e. $f(n) = n$) and the other is the negation map (each number maps to its negative i.e. $f(n) = -n$). This is because an automorphism in this group must preserve the group operation, which is addition, and these two functions are the only ones that do this. So, there are only two homomorphisms from $(Z, +)$ to itself:

The identity mapping (the function that maps every integer to itself), $f(n) = n$, and

The negation function, $f(n) = -n$.

Q5 Text Solution:

The correct answer is **Option D (A-I B-III C-II D-IV)**.

List I	List II
(A) PERT	(I) used for non repetitive jobs.
(B) Optimistic time	(III) The shortest possible time to complete the activity if all goes well.
(C) CPM	(II) Used for repetitive jobs.
(D) Pessimistic time	(IV) The longest time that an activity takes.



- **PERT** is designed for non-repetitive projects, focusing on the sequence of tasks.
- **Optimistic time** estimates the best-case scenario for task completion.
- **CPM** is used for repetitive jobs, emphasizing efficiency.
- **Pessimistic time** reflects the worst-case duration for a task.

Q6 Text Solution:

The correct answer is **Option C (Budgeted cost of work performed ÷ Actual cost of work performed)**.

The question asks which formula calculates the Cost Performance Index (CPI), which is determined by dividing the Budgeted Cost of Work Performed (BCWP) by the Actual Cost of Work Performed (ACWP), providing a measure of cost efficiency in project management.

Q7 Text Solution:

The correct answer is: Option **D**

Software validation

It is a method of checking whether the software product is up to the mark or in other words product has high-level requirements to show that a system both conforms to its specifications and meets the expectations of the system customer. It is the process of examining the validation of a product i.e. it checks what we are establishing in the right product.

Q8 Text Solution:

The correct answer is: Option **A**

A. Both Statement I and Statement II are true.

The question asks what happens when a TCP server listening on a specific port crashes and restarts while a client is connected, specifically whether the server can start listening again and how the client will respond if it sends data after the server has rebooted. The server can listen on the same port after reboot, and if the client sends a packet to the server after it has restarted, it will receive a RST segment because the original connection no longer exists.

Q9 Text Solution:

The correct answer is **D: A is false but R is true.**

Assertion A: "I/O protection is ensured by a hardware trap" — This statement is **false**. I/O protection is generally ensured by the operating system using various mechanisms like privileged modes, rather than just by a hardware trap.

Reason R: "I/O interrupt is caused by conditions like I/O completion and device malfunction occurring within the I/O devices" — This statement is **true**. I/O interrupts are triggered by events such as I/O completion or device malfunctions.

Q10 Text Solution:

The correct answer is **Option C (A-III, B-IV, C-I, D-II)**.

List I	List II
(A) Parallel FFT	(III) $O(\log n)$
(B) Iterative FFT	(IV) $O(n \log n)$
(C) Evaluation of polynomial at n points by Homer method	(I) $O(n^2)$
(D) Product of two polynomials that are represented in point value form	(II) $O(n)$

Q11 Text Solution:

The correct answer is **Option C (A-IV B-III C-I D-II)**.

List I	List II
(A) Bezier curves	(IV) painting drawing CAD.
(B) B-Splines	(III) Bais and tension
(C) Sweep representation	(I) 3 dim objects that have translation and rotational symmetry.
(D) Natural Objects	(II) Fractal geometry.

(A) Bezier curves are mathematical descriptions of curves that can accurately describe various



shapes. They are widely used in computer graphics, including painting, drawing, and Computer-Aided Design (CAD).

(B) B-Splines short for Basis Splines, it provides more control over curves' shapes. The notion of bias and tension are key features of design layouts that B-Splines can often accommodate. They allow for control over how "tight" or "loose" a curve is and in which direction the curve tends to lean.

(C) Sweep representations refer to the method that represents a 3D object as a 2D profile swept along a path. These methods are capable of creating complex 3D objects with translation and rotational symmetry, as the same pattern is repeated in different orientations and positions.

(D) Natural objects: Fractal geometry is a special form of graphical representation of mathematical functions. This helps to explain structures in the world that have complexity, detail, and hierarchy, such as those found in natural landscapes. As a result, fractal geometry has wide-ranging applications in many fields, including the representation of natural objects in computer graphics

Q12 Text Solution:

The correct answer is **Option A (Both Statement I and Statement II are correct.)**.

Statement I: $f(n) \geq 1$ and $f(n) = O(g(n)) \Rightarrow g(n) = \Omega(f(n))$

This statement is correct. If $f(n)$ is a positive function ($f(n) \geq 1$) and $f(n) = O(g(n))$, it means that $g(n)$ grows at least as fast as $f(n)$ for large values of n . This is because $g(n)$ is an upper bound for $f(n)$, and since $f(n)$ is positive, $g(n)$ must also be positive and grow at least as fast as $f(n)$. Therefore, $g(n) = \Omega(f(n))$, which means that $g(n)$ is a lower bound for $f(n)$.

Statement II: $f(n) = O(g(n)) \Rightarrow \lg(f(n)) = O(\lg(g(n)))$ where $\lg(g(n)) \geq 1$ for all sufficiently large n .

This statement is also correct. If $f(n) = O(g(n))$, then it means that $f(n)$ grows at most as fast as

$g(n)$ for large values of n . Taking the logarithm of both sides,

we get $\lg(f(n)) = O(\lg(g(n)))$. Since $\lg(g(n)) \geq 1$ for all sufficiently large n , the logarithmic function is monotonically increasing, and therefore, $\lg(f(n))$ grows at most as fast as $\lg(g(n))$.

Q13 Text Solution:

The correct answer is **Option B (S1, S3 false, and S2 is true)**.

S1: LRU page replacement algorithm does not suffer from Belady's anomaly. Belady's anomaly occurs where increasing the number of page frames results in an increasing number of page faults and this occurs in FIFO (First in First out) page replacement algorithm.

S2: Shortest Remaining Time First (SRTF) scheduling may cause starvation. It can lead to starvation for longer processes.

S3: Stack is not shared by all threads in a process because each thread has its own stack. Some of the resources that threads share within the same process are the code section, data section, memory, file descriptors and processor.

Q14 Text Solution:

The correct answer is **Option B:** $(F + P) \times N$

The expression $(F + P)$ combines the ratio of the size of a data entry to the size of a data record (F) with the average number of records per data page (P). By multiplying this sum by the total number of data pages (N), we account for the overall cost of retrieving records in sorted order from an unclustered B+ tree, reflecting the total overhead associated with both the number of records and their sizes during the traversal process.

Q15 Text Solution:

Option C: A-IV B-III C-I D-II

Explanation:

The correct sequence goes like:

	List		List -
	- I		II
A.	Scenario Testing	IV.	The documentation of a use case.



B.	Regression Testing	III.	No new bugs after changes in program
C.	Component Testing	I.	To verify the I/O behaviour of text object
D.	Beta Testing	II.	user acceptance methodology.

Scenario testing: Software testing methods such as scenario testing validate a system's behavior by mimicking real-world use cases or user interactions.

Regression testing: Regression testing is the process of re-running both functional and non-functional tests to make sure that software that has already been designed and tested continues to function as intended even after modifications or problem patches.

Component testing: The process of testing individual software modules or components separately to make sure they work properly before integrating them into a larger system is known as component testing.

Beta testing: A pre-release version of the application is made available to a select group of users outside the development team during the beta testing phase of software testing in order to collect input, find defects, and evaluate usability in practical settings.

Q16 Text Solution:

Let's analyze the statements step by step:

A. If some NP-complete problem P is in P , then $P=NP$

This is correct. The definition of NP-complete problems implies that if one NP-complete problem can be solved in polynomial time, all problems in NP can also be solved in polynomial time, leading to the conclusion that $P=NP$.

B. TSP (Traveling Salesman Problem) is in NP. This is also correct. The decision version of TSP can be verified quickly (in polynomial time) by

checking a given tour against the specified distance.

C. SAT (Boolean Satisfiability Problem) is in NP. This statement is true as well. SAT is the original NP-complete problem, and its solutions can be verified in polynomial time.

D. The Hamiltonian Circuit problem is not NP-complete.

This is incorrect. The Hamiltonian Circuit problem is indeed NP-complete, meaning it is as hard as the hardest problems in NP.

The only true statements are A, B, and C.

Therefore, the correct answer is A, B, and C only.

Q17 Text Solution:

Explanation:

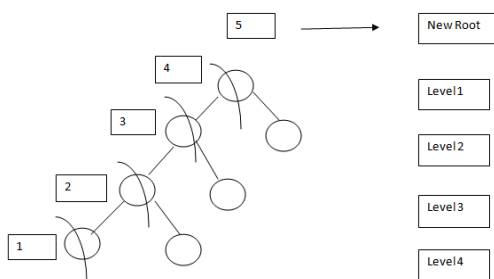
Assume all nodes are filled, which implies that each node holds $n - 1$ keys. Given that, the tree has four levels. If a new key is inserted, a new node is produced at each level. In the worst-case scenario, the root node will be divided into two halves, resulting in four layers. So the answer should be 5 because the tree will be grown by one level.

Steps:

- Initially, all levels and nodes are full of keys.
- When another key arrived, it went below the leaf.
- The overflow occurs on the leaf and it is divided into two sections, and one key is relocated to the top level.
- Again, overflow occurs. That node is divided into two halves, with one key transferred to the top level.
- The same will occur until we reach the root, which is divided into two segments and has a new root.
- So, upon all 'n' levels, we get one more node as well as a new root. So there are a total of $n+1$ additional nodes. Hence the answer is $4+1 = 5$.

Structure of B- tree:



**Q18 Text Solution:**

Correct Option- **A (EMPTY: REAR == FRONT
FULL: (REAR +1) mod n == FRONT)**

Explanation:

To determine the conditions for a circular queue implemented using an array, we define two pointers: **REAR** and **FRONT**.

Conditions for the Queue:

Empty Queue Condition: The queue is considered empty when the two pointers are equal. Thus, we have:

EMPTY: REAR=FRONT

Full Queue Condition: The queue is full when incrementing the REAR pointer by one (considering the circular nature) would cause it to meet the FRONT pointer. This can be expressed as:

FULL: (REAR+1)mod n=FRONT

Q19 Text Solution:

Option - **A (Both Statement I and Statement II are correct)**

Explanation:

To evaluate the statements about the software product line system:

Statement I: This describes that at the interaction level, components provide an operator display interface and communication system interfaces. This is generally accurate since interaction levels typically focus on user interfaces and communication aspects.

Statement II: This states that at the I/O management level, components are responsible for operator authentication, report generation, and query management. This is also correct, as

I/O management usually encompasses these tasks.

Q20 Text Solution:

Option - **B (Both A and R are true but R is NOT the correct explanation of A)**

Explanation: To analyze the statements:

Declaration A Content: To construct a thread, a process uses a library function.

Analysis: This assertion is true. The majority of programming environments use library functions (such as `std::thread` in C++ or `pthread_create` in POSIX threads) to create new threads. A process is able to create and manage threads thanks to this function.

Rationale R Content: To inform the Kernel of their resource and input/output needs, the threads send system calls.

Analysis: This assertion is true as well. Similar to processes, threads can conduct I/O activities and make system calls to request resources. To handle these needs, they communicate with the kernel.

The claim is made that a process can generate threads by utilizing a library function. Although the explanation explains how threads interact with the kernel for resource requirements, it doesn't address the allegation regarding the library function that is utilized to generate threads.

Considering that both claims are true but the explanation does not directly address the claim,

Q21 Text Solution:

Option- **C (A is true but R is false)**

Explanation:

Let us dissect the claims:

Assertion A: Since AVL trees adhere to a more stringent balancing criterion (balance factors of -1, 0, or +1), they are more balanced than Red-Black trees. More balanced trees result from this rigorous balancing, which keeps the AVL tree's height below that of a Red-Black tree. More rotations may be required during insertion and deletion processes, though, if this balance is



maintained.

Reason R: Untrue A Red-Black tree with n nodes has a maximum height of $2 \log_2(n+1)$ meaning that it can be higher than $\log_2 n$ but does not necessarily have to be; for some values of n , it can be equal. The statement for AVL trees is also incorrect; the correct upper bound for the height of an AVL tree with n nodes is approximately $1.44 * \log_2(n+2) - 1.329$.

Conclusion: AVL trees are better balanced and might need more rotations, as stated in Assertion A.

Reason R is untrue because the heights of stringent inequalities are given incorrectly.

Q22 Text Solution:

Option - C (A-III B-I C-II D-IV)

Explanation:

	List - I		List - II
A.	Physical Layer	III.	Transmission of raw bits over communication channel
B.	Datalink layer	I.	Transforming the raw bits in the form of frame for transmission
C.	Network Layer	II.	Control and monitoring of subnet
D.	Transport Layer	IV.	Datagram transmission data through connection-oriented or connectionless using datagram

- 1. Physical Layer:** Handles wires, switches, and electrical impulses that are used to physically connect devices.
- 2. Data Link Layer:** Controls error detection and correction, framing, and dependable communication between devices that are directly connected.
- 3. Network Layer:** Chooses the optimal path for data packet forwarding and routing between devices on various networks.
- 4. Transport Layer:** Oversees end-to-end communication, guarantees full data transfer, and offers flow control and error recovery.

Q23 Text Solution:

Option- A (36)

Explanation:

Locate the Intersection Numbers that are multiples of 3 and 5 to be included in $R \cap S$. Fifteen is the least common multiple (LCM) of three and five.

These are the multiples of 15, 30, 45, 60, 75, and 90 from 15 to 100.

After counting these, we get: $15n \leq 100 \Rightarrow n \leq 6$.

Hence, $R \cap S$ has 6 terms: $|R \cap S| = 6$.

Determine $(R \cap S) \times (S \cap R)$

$|(R \cap S) \times (S \cap R)| = |R \cap S| \times |S \cap R| = 6 \times 6 = 36$ since $R \cap S = S \cap R$.

Q24 Text Solution:

Option - A (gives a lossless join, and is dependency preserving)

Explanation:

Concept 1: Loseless Joining

In given question,

Let $R_1 = (A, B)$, $R_2 = (B, C)$, $R_3 = (B, D)$

- $R_1(A, B) \cup R_2(B, C) \cup R_3(B, D) = R(ABCD)$ (SATISFIED)

- $R_1 \cap R_2 \cap R_3 \neq \emptyset$

- Yes, there is a lossless join because B is a shared attribute across all subrelations, which permits reconstruction without loss.

Concept 2: Dependency Preserving

In the given question, All functional dependencies must be able to be expressed in terms of the decomposed relations for a decomposition to be considered dependency preserving.

In (A, B) : We can maintain $A \rightarrow B$.

In (B, C) : We can maintain $B \rightarrow C$.

In (B, D) : We can maintain $D \rightarrow B$.

Since the sub-relations suggest functional dependencies, the answer is yes.

Conclusion : R gives a lossless join, and is dependency preserving

Q25 Text Solution:

Option - B ($L(F_1) = L(F_2)$)

Explanation:

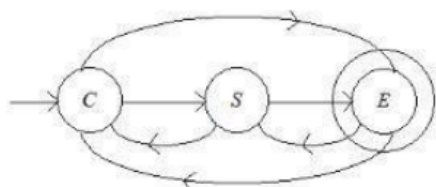


When a finite automaton F1 is flipped to produce F2, the language accepted by F2 is the inverse of that accepted by F1.

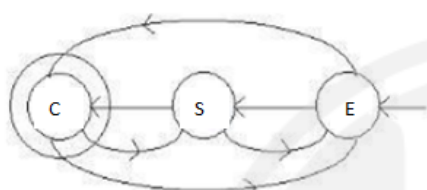
In other words, if F1 accepts language L, F2 will accept the converse.

Thus, $L(F1) = L(F2)$

F1:



F2: Reverse of F1



Q26 Text Solution:

Option- **C (Statement I is correct but Statement II is incorrect)**

Explanation:

Statement I: "Address translation is performed for every logical address used during the execution of a program."

This is **true**. Every logical address that a program generates in a virtual memory system needs to be converted into a physical address in memory. The Memory Management Unit (MMU) performs this translation for each logical address that is accessed.

Statement II: "A program can execute only when all of its components are loaded in the memory." In the context of virtual memory, this is **false**. The ability to run a program even when not all of its components are put into physical memory at once is one of the key advantages of virtual memory. The operating system can swap out other components and only load those that are required.

Q27 Text Solution:

Option - **B ($1/2 MN (M + N - 2)$)**

Explanation:

Given: M points on line AB, N points on line AC.

Valid Combinations for Triangle Formation:

1. Choose 2 points from line AB and 1 point from line AC.
2. Choose 1 point from line AB and 2 points from line AC.

Calculation for every possible pairing:

1. Selecting 2 points from line AB and 1 from line AC

- No. of ways to choose 2 points from M is $= M (M - 1) / 2$
- No. of ways to choose 1 point from N is N.
- Therefore, the total combinations: $T1 = (M (M - 1) / 2) * N$

2. Selecting 1 point from line AB and 2 from line AC

- No. of ways to choose 1 point from M is M.
- No. of ways to choose 2 points from N is $= N (N - 1) / 2$
- Therefore, the total combinations: $T2 = (N (N - 1) / 2) * M$

Total Number of Triangles: The total number of triangles T can be expressed as:

$$T = T1 + T2 = \{ MN (M - 1) / 2 \} + \{ MN (N - 1) / 2 \}$$

Simplifying the Expression:

$$T = MN / 2 ((M - 1) + (N - 1))$$

$$T = 1/2 MN (M + N - 2)$$

Q28 Text Solution:

Option - **A (Both Statement I and Statement II are correct)**

Explanation:

First statement: Fuzzifier is a part of a fuzzy system

This assertion is **true**. Based on membership functions, the fuzzifier transforms crisp inputs into fuzzy values. It is essential to the fuzzy inference procedure.

Second statement : Inference engine is a part of fuzzy system



This is also **true**. After processing the fuzzy rules, the inference engine produces fuzzy outputs based on the fuzzy inputs. It is a key element of a fuzzy system that makes judgments based on the rules established within the system.

Q29 Text Solution:

The correct answer is **A, C and D only**.

A. A database design is in BCNF if each member of the set of relation schemas that constitutes the design is in BCNF. This statement is true because the definition of BCNF (Boyce-code Normal Form) states that for every dependency $X \rightarrow Y$ in a relation. X must be a super key and this applies at schema level (a set of attributes) in a relation. If all relation schemas are in BCNF, the database design can said to be in BCNF.

B. A BCNF schema can have transitive dependency. This statement is false because according to the definition of BCNF for every dependency $X \rightarrow Y$ in a relation. X must be a super key and this disallows transitive dependencies (i.e. if $X \rightarrow Y$ and $Y \rightarrow Z$ then $X \rightarrow Z$) which would imply that there is a non-superkey attribute that determines another attribute.

C. It is always possible to obtain a 3 NF design without sacrificing a lossless join. This statement is true because normalizing a database schema to 3NF can be done without losing the ability to perform lossless join by ensuring that whenever any functional dependency $X \rightarrow Y$ causes a violation, the decomposition of relation includes X in all decomposed relation.

D. There are multivalued dependencies in 4 NF. This statement is true because 4NF (Fourth Normal Form) is designed to handle multivalued dependencies because multivalued dependency always require at least three attributes because it consists of at least two attributes that are dependent on a third.

Bases on the explanation statements **A, C and D are correct**.

Q30 Text Solution:

The correct answer is **Option A (presentation adequacy)**.

A. Presentation adequacy refers how well the representation can convey the intended meaning to a user.

B. Inferential adequacy refers to the representation's ability to support reasoning and inference based on the knowledge it encapsulates.

C. Inferential efficiency relates to how quickly and efficiently the system can perform inference operations and make logical deduction in a timely manner.

D. Acquisitional efficiency is concerned with how well the system can acquire new knowledge or adapt to changes in environment.

Presentation adequacy is not considered a property of a good system for representation of knowledge because it focuses on user experience rather than logical capabilities, unlike inferential adequacy, efficiency properties which are essential for accurate reasoning processes.

Q31 Text Solution:

The correct answer is **24**.

$a = 0x10$ (prefix 0x is used for hexadecimal numbers) = 10 in hexadecimal = 16 in decimal
Hexadecimal to decimal conversion

$$(10)_{16} = 1 \times 16^1 + 0 \times 16^0 = 16 = (16)_{10}$$

$b = 010$ (prefix 0 is used for octal) = 10 in octal = 8 in decimal

Octal to decimal conversion

$$(10)_8 = 1 \times 8^1 + 0 \times 8^0 = 8 = (8)_{10}$$

$$c = a + b = 16 + 8 = 24$$

In printf %d is used to print decimal numbers so the final answer should be in decimal only.

Q32 Text Solution:

The correct answer is **1100.0100**

Fixed point representation is a way of storing real number in binary form. A 4+4 binary fixed point accommodates 4 bits for the integer part (one of them being the sign bit) and 4 bit for fractional part.



Binary equivalent of $-3.72 =$

step1 : Integer part -3

- The binary representation of 3 is 0011.
- In 2's complement system (which is used to represent negatives) flip the bits (1100) and add 1 to it. It gives the representation of -3 is 1101.
- Since we are representing a 4-bit binary number, we need to consider only 4 bits which is 1100 after discarding overflow (overflow is generally considered an error and it is silently ignored is 2's complement binary arithmetic.)
- So, the binary representation of -3 is 1100.

step2 : Fractional part 0.72

Keep multiplying by 2

- $0.72 \times 2 = 1.44 \Rightarrow$ the first digit after the decimal is 1.
- $0.44 \times 2 = 0.88 \Rightarrow$ the second digit after the decimal is 0.
- $0.88 \times 2 = 1.76 \Rightarrow$ the third digit after the decimal is 1.
- $0.76 \times 2 = 1.52 \Rightarrow$ the fourth digit after the decimal is 1.

Here, we get 1011 for the fractional part and the complement of this is 0100.

Hence, The binary equivalent of 0.72 in 4-bit precision is .0100

So, 1100.0100 is the 4+4 bit binary fixed point equivalent of $-(3.72)_{10}$.

Q33 Text Solution:

The correct answer is **option D (An equivalence relation)**.

Equivalence Relation : A relation R on a set A is said to an equivalence relation if it is transitive, symmetric and reflexive.

Step 1: First we will check R for reflexive.

Reflexive Relation : A relation R on a set A is said to be reflexive if $(a,a) \in R, \forall a \in A$.

Given N is a set of natural numbers and R is a relation on $N \times N$ defined by $(a,b)R(c,d)$ if $ad(b+c) = bc(a+d)$.

For, $(a,b)R(a,b)$

$$\Rightarrow ab(b+a) = ba(a+b)$$

$$\Rightarrow ab(b+a) = ab(b+a) \text{ which is true.}$$

So, R is reflexive.

Step 2: Now, we will check R for symmetric.

Symmetric Relation: A relation R on a set A is said to be symmetric if $(a,b) \in R$ then $(b,a) \in R, \forall a, b \in A$.

For, $(a,b)R(c,d)$

$$\Rightarrow ad(b+c) = bc(a+d)$$

$$\Rightarrow bc(a+d) = ad(b+c)$$

$$\Rightarrow cb(d+a) = da(c+b)$$

$$\Rightarrow (c,d)R(a,b) \text{ is true.}$$

So, R is symmetric.

Step 3 : Now, we will check R for transitivity.

Transitive Relation: A relation R on a set A is said to be transitive if $(a,b) \in R$ and $(b,c) \in R$ then $(a,c) \in R, \forall a, b, c \in A$.

For $(a,b)R(c,d)$ and $(c,d)R(e,f)$

$$\Rightarrow ad(b+c) = bc(a+d) \text{ and } cf(d+e) = de(c+f)$$

$$\Rightarrow abd + adc = abc + bcd \text{ and } cfd + cef = ced + def$$

$$\Rightarrow abd - abc = bcd - acd \text{ and } cfd - ced = def - cef$$

$$\Rightarrow ab(d-c) = cd(b-a) \text{ and } cd(f-e) = ef(d-c)$$

$$\Rightarrow (ab)/(b-a) = (cd)/(d-c) \text{ and } (cd)/(d-c) = (ef)/(f-e)$$

$$\Rightarrow (ab)/(b-a) = (ef)/(f-e)$$

$$\Rightarrow abf - abe = efb - efa$$

$$\Rightarrow abf + efa = efb + abe$$

$$\Rightarrow af(b+e) = be(f+a) \text{ So, } (a,b)R(e,f) \text{ is true.}$$

$\therefore$ R is transitive.

As R is reflexive, symmetric, and transitive, R is an equivalence relation.

Q34 Text Solution:

The correct answer is **4 (option C)**.

Context-Free Grammar (CFG) : It is used to describe the syntax or structure of languages. In this CFG, S, A, B, C, D are non-terminal symbols, and a, b, and e are terminal symbols.

These rules show that every non-terminal symbol eventually maps to a single terminal symbol.

$$S \rightarrow AB$$

$$A \rightarrow CD$$

$$B \rightarrow e$$

$$C \rightarrow a$$

$$D \rightarrow b$$



According to the production rule,

$S \rightarrow AB$

$S \rightarrow CDe$ ($A \rightarrow CD$ and $B \rightarrow e$)

$S \rightarrow abe$ ($C \rightarrow a$ and $D \rightarrow b$)

The yielded string has length three. Grammar rule $B \rightarrow e$ is part of CNF grammar and although it appears to produce a single terminal symbol, it is considered to produce two symbols in terms of its yield length. This is because in CNF, each production either produces two non-terminals or one terminal.

S (yield length = 4) $\rightarrow A$ (yield length = 2) + B (yield length = 2)

This sums up to a total yield length of 4.

Therefore, the maximum yield length of this CFG is 4.

Q35 Text Solution:

The correct answer is **option D** $\overline{P \oplus Q}$

XOR operator $\oplus$ is both commutative and associative. Some important properties of XOR

$$A \oplus B = A\bar{B} + \bar{A}B$$

$$A \oplus A = 0$$

$$1 \oplus A = \bar{A}$$

$$0 \oplus A = A$$

$$F(P, Q) = ((1 \oplus P) \oplus (P \oplus Q)) \oplus ((P \oplus Q) \oplus (Q \oplus 0))$$

$$= 1 \oplus (P \oplus P) \oplus Q \oplus P \oplus (Q \oplus Q) \oplus 0$$

$$(A \oplus A = 0)$$

$$= (1 \oplus 0) \oplus Q \oplus P \oplus (0 \oplus 0)$$

$$(0 \oplus A = A \text{ and } A \oplus A = 0)$$

$$= (1 \oplus Q) \oplus (P \oplus 0)$$

$$(1 \oplus A = \bar{A} \text{ and } 0 \oplus A = A)$$

$$= \bar{Q} \oplus P$$

$$(A \oplus B = A\bar{B} + \bar{A}B)$$

$$= \bar{Q} \cdot \bar{P} + Q \cdot P$$

$$= P \odot Q$$

$$(A \odot B = \bar{A} \cdot \bar{B} + A \cdot B)$$

$$= \overline{P \oplus Q}$$

Q36 Text Solution:

The correct answer is **option D** (A transaction reads a data item after it is written by an uncommitted transaction.)

Irrecoverable Error is an error that does not permit the database to return to a consistent state after a failure. They are primarily caused by inconsistencies that occur during transactions.

Option A : A transaction writes a data item after it is read by an uncommitted transaction.

This is known as a Dirty Write. It is not an irrecoverable error, if the uncommitted transaction fails we can use rollback operation to undo the changes.

T ₁	T ₂
R(X)	
	W (X)

Not forms irrecoverable

Option B : A transaction reads a data item after it is read by an uncommitted transaction.

This is not an irrecoverable error. Rollback operations can be implemented to maintain consistency.

T ₁	T ₂
R(X)	
	R(X)

Not forms irrecoverable

Option C : A transaction reads a data item after it is written by a committed transaction.

This is a normal operation within DBMS systems and doesn't lead to an irrecoverable error.

T ₁	T ₂
W (	



X	
)	
c	
o	
m	
m	
it	
	R(
	X)

Not forms irrecoverable

Option D : A transaction reads a data item after it is written by an uncommitted transaction.

This results in a Dirty Read, which can lead to an irrecoverable error under specific circumstances:

T ₁	T ₂
W	
(	
X	
)	
	R(
	X)
	c
	o
	m
	m
	it
a	
b	
o	
r	
t	

Dirty read may cause irrecoverable of T2 when T2 commits before T1.

- If the transaction that issued the read operation commits before the transaction that issued the write operation, and then the system fails which causing the uncommitted transaction to abort.
- In this case, the committed transaction read and used uncommitted data, which can't be rolled back, leading to an irrecoverable state.

Q37 Text Solution:

The correct answer is **Option C (Ethnography)**.

Ethnography is an observational technique that can be used to understand operational processes and help derive requirements for software to support operational processes. Through ethnography, requirements for software can be derived organically from the environment in which it will be used, rather than being imposed based on assumptions or out-of-context criteria.

Q38 Text Solution:

The correct answer is option B. **Both A and R are true but R is NOT the correct explanation of A.**

First In First Out Page Replacement Algorithm :

In this algorithm, the operating system keeps track of all pages in the memory in a queue, oldest page is in the front of the queue. When a page needs to be replaced page in the front of the queue is selected for removal.

FIFO Page replacement algorithms suffer from Belady's anomaly :

Belady's anomaly states that it is possible to have more page faults when increasing the number of page frames.

Statement P: Increasing the number of page frames allocated to a process sometimes increases the page fault rate.

Correct, as FIFO page replacement algorithm suffers from belady's anomaly which states the above statement.

Statement Q: Some programs do not exhibit a locality of reference. Correct, Locality often occurs because code contains loops that tend to reference arrays or other data structures by indices. So we can write a program that does not contain a loop and does not exhibit locality of reference.

So, both statements P and Q are correct but Q is not the reason for P as Belady's Anomaly occurs for some specific patterns of page references.

Q39 Text Solution:

The correct answer is **option B (A - III, B - IV, C - I, D - II)**.

Solution :

List I	List II
--------	---------



A. CDMA	III. Accessing mechanism for multiple transmitters over a single channel
B. GSM	IV. It is cellular technology employs hybrid of FDMA and TDMA
C. UMTS	I. It provides mobile internet connection with faster data transfer rates
D. WiFi	II. It allows users to connect to a network or to other devices over wireless channel

- **CDMA (Code Division Multiple Access)** is cellular technology uses a spread spectrum technique, enabling multiple users to share a single frequency channel. Its primary purpose is to provide mobile voice and data services.
- **GSM (Global System for Mobile Communications)** is a digital cellular technology that helps devices connect to the internet.
- **UMTS (Universal Mobile Telecommunications System)** is a 3G cellular technology providing faster data transfer rates than GSM.
- **Wi-Fi (Wireless Fidelity)** is a wireless networking technology that permits devices to connect to the internet or to each other over a short distance.

Q40 Text Solution:

The correct answer is language is **context-free but not linear (option A)**.

Solution :

$L = w \in \{a,b,c\}^* : n_a(w) + n_b(w) = n_c(w)$. This language is context-free but not linear.

Example :

$S \rightarrow aSc$ (This rule adds one 'a' and one 'c' to string and maintains balance.)

$S \rightarrow bSc$ (This rule adds one 'b' and one 'c' to string and maintains balance.)

$S \rightarrow \epsilon$ (This rule generates empty string.)

Using this CFG, you can generate strings that satisfies the above condition.

$S \rightarrow aSc$ ($S \rightarrow bSc$)

$S \rightarrow abSc$ ($S \rightarrow \epsilon$)

$S \rightarrow abcc$

So, the language is context-free but not linear.

Q41 Text Solution:

The correct answer is **Difference (option D)**.

Mutation operators in a Genetic Algorithm is a process that modifies the genes of a chromosomes to introduce variation into the population.

A. Random Resetting: This is a type of mutation operator where a specific gene in a chromosome is randomly chosen and assigned a new value from the feasible parameter range.

B. Scramble: This operator randomly selects two points in the chromosome and then shuffles the genes between those points.

C. Inversion: Inversion mutation operators select a substring (i.e., a portion of the chromosome) and revert the order of genes within that.

D. Difference: This is not a known mutation operator in the context of genetic algorithms.

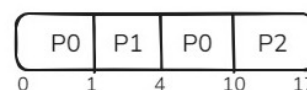
Q42 Text Solution:

The correct answer is **3.67ms (option B)**.

In **Shortest Job First (SJF) Scheduling**, the process with the smallest burst time is processed next.

In **Preemptive SJF**, if a new process arrives with CPU burst length less than remaining time of current executing process, CPU is preempted and that new process starts execution.

Gantt Chart



Processes	Arrival Time (A.T.)	Burst Time (B.T.)	Completion Time (C.T.)	Turn Around Time (T.A.T.) = (C.T. - A.T.)	Waiting Time (T.A.T. - B.T.)
P0	0	7	10	10	3
P1	1	3	4	3	0



P2	2	7	17	15	8
----	---	---	----	----	---

So, the average waiting time = $(3\text{ms} + 0\text{ms} + 8\text{ms}) / 3 =$

3.67ms

=> **3.76 ms** is the average waiting time for the three processes.

Q43 Text Solution:

Option - D (Meaning Analysis)

Meaning analysis is not the component of the natural language understanding process.

Natural Language Understanding Process: The umbrella of natural language processing in artificial intelligence includes natural language understanding and interpretation. Numerous uses for it exist, including digital assistants, chatbots, as and automatic translation services.

There are five components/ phases of NLP are as:

Step 1: Lexical Analysis and

Morphological: Lexical analysis is the first stage of NLP in which the source code is scanned as a character stream and transformed into meaningful lexemes. It breaks the text into words, phrases, and paragraphs.

Step 2: Syntactic Analysis: Syntactic analysis checks grammar, arranges words, and illustrates word relationships.

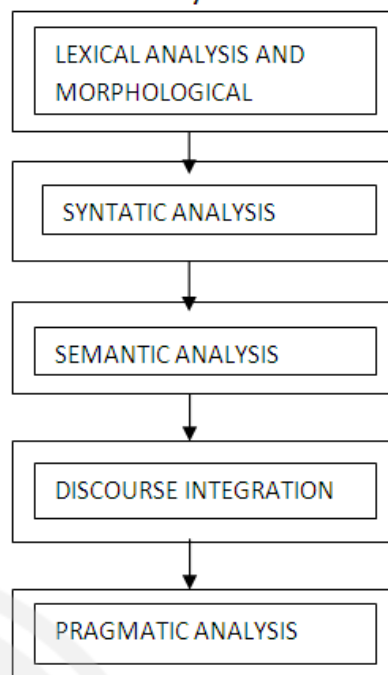
Step 3: Semantic Analysis: Semantic analysis is focused on the representation of meaning. It primarily concentrates on a word, phrase, or sentence's literal meaning.

Step 4: Discourse Integration: Discourse integration draws meaning from the sentences that come before it as well as from those that follow.

Step 5: Pragmatic Analysis: The fifth and final stage of NLP is pragmatic. It assists you in determining the desired outcome by using a set

of guidelines typical of cooperative talks.

COMPONENTS/PHASES OF NLP:



Q44 Text Solution:

Option 4: _1,3,8,10,5_

CONCEPT

Hashing: The process of dividing a big volume of data into manageable tables using an algorithm is known as hashing.

Closed Hashing or Open Addressing:

- All keys are stored solely within the hash table.
- The hash table's size cannot be exceeded by the amount of keys that must be kept inside the hash table.

Linear Probing: If a collision happens during the allocation of a location to a key value in linear probing, we proceed to linearly probe towards the next empty location and assign it to that key.

EXPLANATION: In order to insert the given sequence to the hash table with the help of closed hashing, we make use of the given hash function $h(x) = (6x + 3) \bmod 4$.

Now, going through each element one by one as given in the question.

Table size is seven and it starts from index 0.

Initially, table looks like:

0	1	2	3	4	5	6



Step 1: Insert 1:

Hash Value = $(6 * 1 + 3) \bmod 4 = 9 \bmod 4 = 1$

1 will go to index 1. Table will look like:

0	1	2	3	4	5	6
	1					

Step 2: Insert 3:

Hash Value = $(6 * 3 + 3) \bmod 4 = 21 \bmod 4 = 1$

1 will go to index 2. Table will look like:

Collision occurs. Following linear probing, the table becomes:

0	1	2	3	4	5	6
	1	3				

Step 3: Insert 8:

Hash Value = $(6 * 8 + 3) \bmod 4 = 51 \bmod 4 = 3$

8 will go to index 3. Table will look like:

0	1	2	3	4	5	6
	1	3	8			

Step 4: Insert 10:

Hash Value = $(6 * 10 + 3) \bmod 4 = 63 \bmod 4 = 3$

10 will go to index 4 due to linear probing. Table will look like:

0	1	2	3	4	5	6
	1	3	8	10		

Step 5: Insert 5:

Hash Value = $(6 * 5 + 3) \bmod 4 = 33 \bmod 4 = 1$

5 will go to index 5 due to linear probing. Table will look like:

0	1	2	3	4	5	6
	1	3	8	10	5	

Q45 Text Solution:

Option - B ($\&(A[0][0][0] + w(b * c * r + c * s + t))$)

Explanation:

In C++, memory is allocated for a three-dimensional array expressed as `int A[a][b][c]` in a pattern beginning with 'c', followed by 'b', and ending with 'a'. It's so because the items are arranged in row-major order. Thus, while attempting to calculate an element's address `A[r][s][t]`,

- Since we're traveling in the 'c' dimension, 't' will certainly have a direct impact on its value.

- 's' must account for all of the 'c' steps required for each shift in the 'b' direction; therefore, 's' contributes to 'c*s' shifts.

- 'r' will have to consider all 'bc' steps we have to take for every shift in the 'a' direction, hence 'r' contributes to 'bc*r' shifts.

Q46 Text Solution:

Option - B

The correct sequence is:

A. Raid- 2 III. Error Correcting parity

B. Raid- 4 IV. Block - interleaved parity

C. Raid- 5 II. Block - interleaved distributed parity

D. Raid- 3 I. Bit - interleaved parity

Below table explains data distribution in RAID:

Raid - 0	Striping
Raid - 1	Mirroring
Raid - 2	Hamming Code
Raid - 3	Bit- interleaved parity
Raid - 4	Block- interleaved parity
Raid - 5	Block- interleaved distributed parity
Raid - 6	Block - interleaved dual distributed parity

Q47 Text Solution:

Option - C (A-IV B-III C-II D-I)

	List- I		List-II
A.	Expert System	IV.	Mycin
B.	Fuzzy System	III	Inference engine
C.	Operator in genetic algorithm	II.	Scramble
D.	Supervised technique	I.	Decision tree

Explanation-

Option A: Expert system- IV. Mycin



- Mycin is an example of expert system designed for medical diagnosis.

Option B: Fuzzy system- III. Inference Engine

- The inference engine is a component of a fuzzy system that handles fuzzy logic rules.

Option C: Operator in the genetic algorithm - II. Scramble

- Scramble is a type of cross-over operator in a genetic algorithm.

Option D: Supervised technique - I. Decision tree

- Decision trees are a type of supervised learning technique for both classification and regression.

Thus option, C is true.

Q48 Text Solution:

Option - C (D only)

Explanation-

Option A: LR (0) grammar generates context-free languages that are accepted by deterministic pushdown automata and have the prefix property. This assertion is erroneous. While LR (0) grammar generates context-free languages, they are not required to be accepted by deterministic pushdown automata.

Option B: If M1 is a single tape TM mimicking a multitape TMM, the time needed to simulate n movements is (n^3). This assertion is erroneous since the time spent by the single tape TM, M1, is quadratic, i.e. $O(n^2)$.

Option C: When a pushdown automaton has only one auxiliary memory, it acts similarly to a turning machine. This assertion is erroneous since the PDA with two or more stacks does not have the same processing capacity as Turing machines.

Option D: $L = \{a^n b^n c^n : n \geq 1\}$ is context-sensitive, rather than context-free. This assertion is accurate since $L = \{a^n b^n c^n : n \geq 1\}$ is context-sensitive, but not context-free.

Note: Options are given wrong in the UGC NET answer key. We have updated option C

Q49 Text Solution:

Option - B (4 megabits per second)

Explanation -

1. The first thing we must do is figure out how much data is transferred in a single transaction.

An article typically consists of 3k bytes = 3000 bytes $\times$ 8 bits per byte = 24000 bits.

2. Calculate data sent:

Every two minutes, each of the 20,000 users submits a request.

Consequently, 20,000 users / 2 = 10,000 (transactions will occur in a minute)

3. The total quantity of data exchanged in a minute.

10,000 transactions $\times$ 24,000 bits/transaction = 240×10^6 bits

We must convert this amount because the enquiry asks for throughput in megabits per second.

60 seconds is one minute, so the throughput is 240

Throughput = 240×10^6 bits/minute / 60 seconds/minute = **4 megabits/second**

Q50 Text Solution:

Option- B (2600)

Explanation - Given question is an application of integer partition and combination.

Trick- Shift everything to left to make all variables unconstrained.

Step 1- Subtract constraint from each variable

$$x' = x - 1$$

$$y' = y - 2$$

$$z' = z - 3$$

$$w' = w - 0$$

Step 2- Express equation in terms of new variables:

$$x' + y' + z' + w' = 29 - (1+2+3+0) = 23$$

The basic formula for combinations with repetitions is $C(n+r-1, r-1)$

where, n - sum

r- Number of variables



Therefore, $C(23 + 4 - 1, 4 - 1) = C(26, 3)$
 $26! / ((26 - 3)! * 3!) = 2600$

Q51 Text Solution:

Option B: EC

Explanation: Given, $R(A, B, C, D, F)$ as a relational schema with following functional dependencies:

$C \rightarrow F, E \rightarrow A, EC \rightarrow D, A \rightarrow B.$

We have to find the key for R

Taking closure of the given options we found:

A. $[CD] = [CDF]$

B. $[EC] = [ABCDEF] = [ABCDF] = R$ (Given Relation)

C. $[AE] = [ABE]$

D. $[AC] = [ABCF]$

Looking at the closure of EC, we found it to be equal to R.

Q52 Text Solution:

Option - B (Nagpur Nagpur Nagpur)

Explanation - In C programming, Struct is a datatype which is used for packaging data where each member has different type.

In the given code,

ptr -> z point to Nagpur as it was initialized to first structure in array.

a[0].z points to Nagpur

a[2].p -> z point to Nagpur as third structure was initialized to point to the first structure in array.

Thus, output of the code: **Nagpur Nagpur Nagpur.**

Q53 Text Solution:

Option - 4 (Deadlock freedom)

Explanation: ACID properties of transaction are:

A- Atomicity

- Either nothing happens at all or the complete transaction occurs.

C- Consistency

- Both before and after the transaction, the database must be consistent

I- Isolation

- Independent transactions take place without any influence.

D- Durability

Even in the event of a system breakdown, successful transactions continue to occur.

Q54 Text Solution:

Option: B (A-II B-I C-III D-IV)

A. Critical region - II. Mutual Exclusion

- The Critical region is a code segment where the mutual exclusion principle is followed so as to avoid race conditions.

B. Wait/Signal - I. Hoare's Monitor

- The monitor is a synchronization construct that allows all threads to have the ability to block or wait signal.

C. Working Set - III. Principle of Locality

- The working set of page references works on the principle of locality.

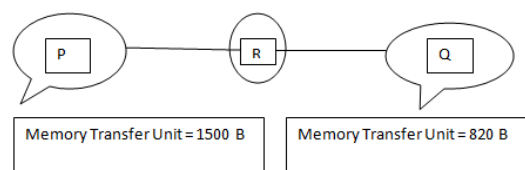
D. Deadlock - IV. Circular Wait

- Circular wait is one of the primary conditions of deadlock.

Q55 Text Solution:

Option- B (A and C only)

Explanation- Given,



Given, TCP Segment size = 1400

Packet Size = 1400 + 20 (Segment Size + IP Header Size) = 1420, The packet will be fragmented at R.

Thus, MTU = 800 + 20 (Header size), TCP segment size will be divided in two parts - 800 and 600

First Fragment = (800 + 20) Second

Fragment = (600 + 20)



Statement A: Two fragments are created at R and the IP datagram size carrying the second fragment is 620 bytes. This is the correct option.

Statement B: If the second fragment is lost, then R resends the fragment with IP identification value of 0x1234

This is an incorrect option because if any one of the fragments is lost then the complete TCP segment has to be retransmitted.

Statement C: If the second fragment lost, then P requires to resend the entire TCP segment.

This is the correct option because if any one of the fragments is lost, then the entire TCP has to be re-transmitted.

Statement D: The TCP destination port can be determined by analyzing the second fragment only.

This is an incorrect option as the destination port number can be determined by looking at any packet or fragment and not just the second frame only.

Q56 Text Solution:

Option - C (A is true but R is false)

Explanation- Dendral is an expert system. Using mass spectrometry data, a novel artificial intelligence project was developed in the 1960s and 1970s to automatically determine the structure of organic molecules. Therefore, claim A is true.

Reason R is untrue. An AI agent's response to its surroundings holds significant influence over its rationality.

Q57 Text Solution:

The probability mass function (PMF) is given by:

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

X represents the number of requests.

λ is the average number of occurrences in a given time period.

On average, there are 20 requests per hour.

45 minutes is: $\frac{45}{60} = 0.75$ hours

Therefore, the average number of requests in 45 minutes is:

$$\lambda = 20 \times 0.75 = 15$$

Now, to find the probability of no requests (i.e., $X = 0$), we use the PMF formula:

$$P(X = 0) = \frac{15^0 \cdot e^{-15}}{0!} = e^{-15}$$

So the correct answer is e^{-15} .

Q58 Text Solution:

Given:

Initial value of semaphore = 7

Number of p(wait) operations = 20

Final value of semaphore = 5

- Each P operation decreases the semaphore by 1.

$$7 - 20 = -13$$

Each V operation increases the semaphore by 1

$$-13 + x = 5$$

$$x = 18$$

So the final answer is option C

Q59 Text Solution:

- Binary Valued: Solutions are encoded as binary strings(0s and 1s)
- Real valued: Solutions are represented using real numbers.
- Permutation: Solutions are represented as ordered sequences of elements.

Combinations are not typically used as a standard representation in genetic algorithms.

So the correct answer is Option D

Q60 Text Solution:

(D) Syntax analysis

Explanation :

(A) Code Optimization: It is responsible for making code faster and smaller. It focuses on optimizing performance and reducing resource utilization after the code has already been verified and processed.

(B) Semantic Analysis: It ensures that the program is semantically correct and makes logical sense. It verifies things like type consistency, variable declarations, and scope,



making sure that operations in the program are meaningful and not just syntactically correct.

(C) Code Generators: In this phase, the compiler converts the optimized intermediate code into machine code (binary instructions) or assembly language. It doesn't deal with syntax or grammar but focuses on generating the final executable code that the CPU can understand.

(D) Syntax Analysis: It is also known as parsing.

This phase is responsible for detecting syntax errors. The source code is checked against the grammar of the programming language. It generates a parse tree or abstract syntax tree (AST) to do this process.

Q61 Text Solution:

(C) In a row of relational table, an attribute can have more than one value

Explanation :

(A) An attribute of an entity can have more than one value:

This is true in the context of the ER diagram.

These are called multivalued attributes in ER models. For example, an entity "Person" could have multiple phone numbers.

(B) An attribute of an entity can be composite:

This is true. Composite attributes are attributes that can be divided into subparts. For example, "Address" could be divided into "Street", "City", and "Postal Code."

(C) In a row of a relational table, an attribute can have more than one value:

This is false. In the relational model, a relational table follows the **first normal form (1NF)**, which means that **each attribute in a table row must have atomic (single) values**, meaning one value per attribute per row.

(D) In a row of a relational table, an attribute can have exactly one value or a NULL value:

This is true. An attribute in a relational table cannot be multivalued, so it has to be exactly one valued. In case of no data, it is set as NULL.

Q62 Text Solution:

(C) 1.18

Max speed up after redesign is calculated by

Amdahl's law :

$$Speedup = \frac{1}{(1-f) + \frac{f}{s}}$$

where, f = fraction of execution time affected by the improvement, i.e. f = 30% = 0.30

S = speedup of the affected part (in this case, the floating-point adder is made twice as fast, i.e. S = 2

$$Speedup = \frac{1}{(1-0.30) + \frac{0.30}{2}}$$

$$Speedup = \frac{1}{0.70 + 0.15}$$

$$Speedup = \frac{1}{0.85}$$

$$Speedup \approx 1.176 \approx 1.18$$

So, the **maximum possible speedup** by this redesign is approximately **1.18**.

Q63 Text Solution:

Statement I: Two-phase locking (2PL)

- Two-phase locking ensures conflict serializability by dividing the transaction into two phases: the growing phase (where locks are acquired) and the shrinking phase (where locks are released). However, it does not guarantee freedom from deadlock.

Statement II: Timestamp ordering

- Timestamp ordering ensures conflict serializability by ordering transactions based on their timestamps. It also guarantees freedom from deadlock because transactions are ordered chronologically, preventing circular wait conditions.

Given this analysis, the most appropriate answer is:

D. Statement I is incorrect, but Statement II is correct.

Q64 Text Solution:

Let's analyse each statement:

A. C-Fuzzy means clustering is a supervised learning method - This is incorrect. C-Fuzzy means clustering is an unsupervised learning method.



B. PCA is used for dimension reduction - This is correct. Principal Component Analysis (PCA) is a technique used to reduce the dimensionality of data.

C. Apriori is not a supervised technique - This is correct. Apriori is an unsupervised learning algorithm used for association rule mining.

D. When a machine learning model becomes so specially tuned to the exact input data that it fails to generalise to other similar data, it is called underfitting - This is incorrect. The correct term for this scenario is overfitting, not underfitting. Based on this analysis, the correct answer is: B and C (Option B).

Q65 Text Solution:

(C) Convex polygons

The Sutherland–Hodgman algorithm is an algorithm used for clipping polygons. It works by extending each line of the convex clip polygon in turn and selecting only vertices from the subject polygon that are on the visible side.

Q66 Text Solution:

Explanation:

A. The set of Turing machine codes for TMs that accept all palindromes (possibly along with some other inputs) is decidable.

False — Determining whether a TM accepts all palindromes is an undecidable problem (since it's a property of the language recognized by the TM).

B. The language of codes for TMs that, when started on a blank tape, eventually write a 1 somewhere on the tape is undecidable.

True — This is similar to the Halting Problem; it's undecidable to check if a TM will ever write 1.

C. The language accepted by a Turing machine M (i.e., $L(M)$) is always recursive.

False — The language accepted by a TM may be recursively enumerable (RE) but not necessarily recursive.

D. Post's Correspondence Problem is undecidable.

True — This is a well-known undecidable problem in computability theory.

Correct Answer: Option D — B and D only

Q67 Text Solution:

Option Analysis:

Not a homomorphism: A homomorphism preserves the group operation. In this case, $\phi(xy) = \phi(x)\phi(y)$, which means it does maintain the multiplication operation, so this option is wrong.

One-to-one homomorphism but not onto: One-to-one means different elements in G map to different values in H . However, many different positive numbers in G all map to 1, and many negative numbers map to -1. So, this option is also wrong since it's not one-to-one.

Onto homomorphism but not one-to-one: The map ϕ covers all values in H (both -1 and 1), so it's onto. But it's not one-to-one because many numbers from G map to the same value in H . This option is correct.

Just a homomorphism: While true, this option doesn't fully describe the nature of the map. It's a homomorphism, but more specifically, it's onto and not one-to-one.

Conclusion:

The correct choice is **C** because the map is onto but not one-to-one.

Q68 Text Solution:

Operator Precedence: An operator precedence parser is a bottom-up parser that interprets an operator grammar. This parser is only used for operator grammars.

SLR (Simple LR): This is an efficient LR parser that can work with many different grammars. Still, it's not as powerful as the canonical LR parser.

Canonical LR: This is the strongest type of LR parser and it can handle any context-free grammar without restrictions. It effectively deals with complex and ambiguous situations.

LALR (Look-Ahead LR): This parser improves upon SLR by using more information for better decisions. While it's better than SLR, it isn't as powerful as the canonical LR parser



The correct choice is **C: Canonical LR**

Q69 Text Solution:

In the given C program, the declaration of x is quite complex and involves function pointers and array pointers. Let's break it down step by step to understand it.

Understanding the Declaration:

```
typedef (*(arrfptr[3])()) [10];
```

```
arrfptr x;
```

Step 1: arrfptr[3]

This is an array of three elements ([3])

Step 2: (*(arrfptr[3])())

Each element of the array (arrfptr[i]) is a pointer to a function. The function takes no parameters (()) and returns a pointer.

Step 3: (*(arrfptr[3])()) [10]

The function that each pointer points to returns a pointer to an array of 10 elements ([10]).

Step 4: typedef

This typedef creates an alias arrfptr for this array of pointers to functions.

What is x?

When x is declared as:

```
arrfptr x;
```

It means x is of type arrfptr, which is an array of three function pointers. Each function returns a pointer to an array of 10 elements.

Correct Answer:

Option C: x is an array of three function pointers.

Q70 Text Solution:

Step-by-Step Explanation:

Step 1: Determine the total number of possible numbers

Since we are using 4 digits (1, 2, 4, and 6) and arranging them into four-digit numbers, the total number of such numbers is:

$4! = 24$ (4 factorial means there are 24 different ways to arrange 4 digits).

So, we can make 24 numbers in total.

Step 2: Analyze the contribution of each digit

Every digit (1, 2, 4, and 6) will appear equally in each position (thousands, hundreds, tens, ones).

Since we are making 24 numbers, and there are 4

positions, each digit will appear:

$24/4 = 6$ times in each position.

For example, the digit 1 will appear in the thousands place 6 times, the hundreds place 6 times, and so on.

Step 3: Calculating each digit contributes to the total sum

When a digit appears in the thousands place, its value is multiplied by 1000.

When a digit appears in the hundreds place, its value is multiplied by 100.

When a digit appears in the tens place, its value is multiplied by 10.

When a digit appears in the ones place, its value stays the same.

So, the contribution from each digit to the final sum is based on the place it occupies.

The sum of the digits: The sum of the digits 1, 2, 4, and 6 is:

$$1+2+4+6=13$$

This total sum (13) is the basis for calculating how much each place contributes to the overall total.

Step 4: Calculating the contribution to each place:

Since each digit appears 6 times in each place (thousands, hundreds, tens, ones), the contribution for each place is:

$$6 \times (1000 + 100 + 10 + 1) = 6 \times 1111 = 6666$$

Now, multiply this by the sum of the digits:

$$13 \times 6666 = 86,658$$

Final Answer:

The total sum of all four-digit numbers formed using the digits 1, 2, 4, and 6 is 86,658.

Thus, the correct answer is Option A: 86,658.

Q71 Text Solution:

Option A: Dynamic metrics are collected by measurements made of a program in execution

True: Dynamic metrics involve observing a program's behavior during its execution. This includes performance metrics, memory usage, response time, etc.

Option B: Static metrics are collected by measurements made of representations of the system



True: Static metrics are collected by measurements made of representations of the system like Source code or any other software representation measured without running the program. Code structure metrics and lines of code are a few examples of it.

Option C: The assessment of software quality is an objective process

False: While some parts of software quality, like finding bugs or tracking performance, can be measured with data, other parts are more subjective like usability, and readability of code.

Option D: An important part of quality assurance is the selection of standards that should apply to the software development process

True: Ensuring good software quality means following certain standards throughout development. These standards help keep the process consistent and reliable.

Thus, the correct answer is (D) i.e. A, B, and D.

Q72 Text Solution:

To find, how reliable the system is for an hour of operation using an exponential model, we must first figure out its failure rate (λ).

$$R(t) = e^{-\lambda t}$$

$R(t)$ is the reliability at time t ,

λ is the failure rate

In this case, we are given the time between failures of software:

$$\text{Time between failures} = [6, 4, 8, 5, 6]$$

$$\lambda = 1/\text{Mean time between failures}$$

Mean time between failures (MTBF) is calculated as the average of the time between failures:

$$\text{MTBF} = 6+4+8+5+6/5 = 29/5$$

$$\text{The failure rate } (\lambda) \text{ is } 5/29$$

Put that value in the reliability function:

$$R(t) = e^{-5/29t}$$

For one hour of operation ($t=1$), the reliability is

$$R(1) = e^{-5/29 \times 1} = e^{-5/29}$$

So the correct number is $e^{-5/29}$

Option 3 is correct.

Q73 Text Solution:

Option A: Ackermann's function is primitive recursive.

False: Ackermann's function is an example of a function that grows faster than any simple recursive function. Although it's still computable, it's more complex than basic recursive functions and belongs to a higher class called general recursive functions.

Option B: $L = \{a^n b^k c^{n+k} \mid n \geq 0, k \geq 0\}$ is a regular language.

False: The language $L = \{a^n b^k c^{n+k} \mid n \geq 0, k \geq 0\}$ is **not a regular language**. A finite state machine can't handle counting a's, b's, and c's because it lacks memory. For this, a more powerful machine, like one for context-free languages, is needed.

Option C: $L = \{a^n b^j \mid n = j^2\}$ is not a context-free language.

True: The language $\{a^n b^j \mid n = j^2\}$ is **not context-free**. Context-free grammars (CFGs) can't handle quadratic relationships between variables, like when one variable depends on the square of another. This relationship is too complex for CFGs, so you need context-sensitive grammar (CSGs) to represent it properly.

Option D: For every context-sensitive language L not including λ , there exists some linear bounded automata M such that $L \neq L(M)$

False: For every context-sensitive language L , there is a linear bounded automaton (LBA) M that recognizes it, meaning L is the same as $L(M)$. Therefore, saying that $L \neq L(M)$ is incorrect, as they are equivalent by definition.

Option C is correct.

Q74 Text Solution:

The memory the application has allocated should be freed using the 'free' function from the C standard library. Loop over all rows and free each one before freeing the p itself to release the complete array of pointers allocated memory of 'p'. This however has just allocated the array of pointers and not individual rows.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```



```
#define MAXROW 3
#define MAXCOL 4
int main() {
    int **p, i, j;
    p = (int **)malloc(MAXROW * sizeof(int*));
    // Normally, you would allocate memory for
    each row here.
    // For example:
    // for (i = 0; i < MAXROW; i++) {
    //     p[i] = (int *)malloc(MAXCOL * sizeof(int));
    // }
    // To free the allocated memory, you would
    normally do:
    // for (i = 0; i < MAXROW; i++) {
    //     free(p[i]);
    // }
    free(p); // Free the array of pointers
    return 0;
}
```

So, the correct option is:

D) free(p)

Q75 Text Solution:

Option A: Rule-based preference

Correct: This method is effective because the system resolves conflicts by giving priority to the rule that is triggered first. It guarantees that the most relevant rule is applied when several rules would fit.

Option B: Object-based preference

Correct: This method is good when particular objects hold greater priority. The system organizes and assigns priority to rules based on the involved object, which is beneficial when objects possess specific characteristics that require focus.

Option C: Action-based (rule) preference

Correct: It is reasonable to prioritize according to the action a rule will execute, particularly when the system must assess the implications of various actions. The outcomes of the action influence which rule is deemed more critical.

Option D: Action-based (object) preference

Incorrect: Objects do not carry out actions; rules do. Therefore, assigning priority based on an

"object's action" lacks logic and is not applicable in production systems.

Summary:

A, B, and C are valid approaches because they prioritize based on the rules.

Option D is not correct because objects do not carry out actions. Therefore, it is incorrect to prioritize based on an action performed by an object.

Q76 Text Solution:

Matching the correct expressions:

A. $A \Delta B$ is the symmetric difference of sets A and B, which is defined as $(A-B) \cup (B-A)$. Hence, **A matches with III.**

B. $A - (B \cup C)$ is the set difference, meaning elements of A that are not in $B \cup C$. This is represented by $(A-B) \cap (A-C)$, so **B matches with II.**

C. $A - (B \cap C)$ represents the elements in A that are not at the intersection of B and C. This is equivalent to $(A-B) \cup (A-C)$, so **C matches with I.**

D. $A \cap (B - C)$ represents the elements common to A and $B - C$, which can be expressed as $(A \cap B) - (A \cap C)$. Thus, **D matches with IV.**

Answer:

The correct matching is A where **A - III, B - II, C - I, D - IV.**

Q77 Text Solution:

In computer graphics, "clipping" is a technique that restricts the display of forms and objects to a specified area known as the "clipping window" or "clipping region." Ensuring that only the parts of objects within the visible region are processed and shown, improves both performance and visual clarity.

Clipping is primarily used to "remove" objects and lines from the displayed result by excluding areas of graphics that are outside the clipping region.



Q78 Text Solution:

Using the Shortest Seek Time First (SSTF) scheduling algorithm, we can perform the following calculations to determine the total time required to fulfill all requests:

Initial Position: Cylinder 50 is where the head is located.

Requests: 4, 34, 10, 7, 19, 73, 2, 15, 6, 20 are in order.

Time Calculation: We will move the head to the closest request for each step, figure out how far it is, and add that time to the total.

Step-by-Step Calculation

1. Current Position: 50

- Nearest request: **34** (distance = $50 - 34 = 16$)
- Move to 34: $1 \text{ ms} * 16 = 16 \text{ ms}$
- Total time = 16 ms

2. Current Position: 34

- Nearest request: **20** (distance = $34 - 20 = 14$)
- Move to 20: $1 \text{ ms} * 14 = 14 \text{ ms}$
- Total time = 30 ms

3. Current Position: 20

- Nearest request: **19** (distance = $20 - 19 = 1$)
- Move to 19: $1 \text{ ms} * 1 = 1 \text{ ms}$
- Total time = 31 ms

4. Current Position: 19

- Nearest request: **15** (distance = $19 - 15 = 4$)
- Move to 15: $1 \text{ ms} * 4 = 4 \text{ ms}$
- Total time = 35 ms

5. Current Position: 15

- Nearest request: **10** (distance = $15 - 10 = 5$)
- Move to 10: $1 \text{ ms} * 5 = 5 \text{ ms}$
- Total time = 40 ms

6. Current Position: 10

- Nearest request: **7** (distance = $10 - 7 = 3$)
- Move to 7: $1 \text{ ms} * 3 = 3 \text{ ms}$
- Total time = 43 ms

7. Current Position: 7

- Nearest request: **6** (distance = $7 - 6 = 1$)
- Move to 6: $1 \text{ ms} * 1 = 1 \text{ ms}$
- Total time = 44 ms

8. Current Position: 6

- Nearest request: **4** (distance = $6 - 4 = 2$)
- Move to 4: $1 \text{ ms} * 2 = 2 \text{ ms}$
- Total time = 46 ms

9. Current Position: 4

- Nearest request: **2** (distance = $4 - 2 = 2$)
- Move to 2: $1 \text{ ms} * 2 = 2 \text{ ms}$
- Total time = 48 ms

10. Current Position: 2

- Nearest request: **73** (distance = $73 - 2 = 71$)
- Move to 73: $1 \text{ ms} * 71 = 71 \text{ ms}$
- Total time = 119 ms

Thus, the total time taken to satisfy all requests is **119 ms**.

Q79 Text Solution:

We are given a rectangle with vertices (0,0), (0,2), (3,0), (3,2). We need to apply a scaling transformation to this rectangle, scaling by a factor of 2 towards the x-axis and by a factor of 3 towards the y-axis.

Step 1: Comprehend the scaling transition

Scaling by a factor of 2 along the x-axis entails multiplying the x-coordinates of all points by 2. Scaling by a factor of 3 along the y-axis indicates that the y-coordinates of all points are multiplied by 3.

Step 2: Apply the scaling to each vertex

1. Vertex (0,0):

After scaling, the new coordinates will be: $(0 \times 2, 0 \times 3) = (0,0)$

2. Vertex (0,2):

After scaling, the new coordinates will be: $(0 \times 2, 2 \times 3) = (0,6)$

3. Vertex (3,0):

After scaling, the new coordinates will be: $(3 \times 2, 0 \times 3) = (6,0)$

4. Vertex (3,2)



After scaling, the new coordinates will be: $(3 \times 2, 2 \times 3) = (6, 6)$

Step 3: Conclusion

The new vertices of the rectangle after scaling are $(0, 0)$, $(0, 6)$, $(6, 0)$, $(6, 6)$.

Option C is correct.

Q80 Text Solution:

In computer graphics, scaling is a fundamental change that makes it possible to resize objects while preserving their basic geometry.

This transformation alters an object's size by a scaling factor. It can change the object's size without changing its proportions. There are two types of scaling: non-uniform (different factors for the x and y dimensions) and uniform (same factor for both dimensions).

Q81 Text Solution:

Transmission Delay = Frame Size / Transmission Rate

$$= 512 \text{ bytes} / 10 \text{ Mbps} = 51.2 \mu\text{s}$$

Propagation Delay = Length of medium / Propagation Delay

$$= 2500 \text{ m} / 2 \times 10^8 \text{ m/s} \\ = 12.5 \mu\text{s}$$

Final Conclusion:

Option B is correct.

Q82 Text Solution:

Option A: The identity is unique in any monoid.

True: In any monoid, we have at most one identity element. This characteristic allows everything else in the set to stay the same when combined with it.

Example: Consider the set of non-negative integers $N = \{0, 1, 2, 3, \dots\}$ under the operation of addition (+).

The identity element in this monoid is 0 because, for any $a \in N$, we have: $a + 0 = 0 + a = a$

So, the identity element (0) is unique.

Option B: A monoid is a group if there exists an inverse of each element of the monoid.

True: A monoid satisfies the group property as well, apart from having an identity element, any element has its inverse associated with it.

Example: Consider the set of integers Z under addition.

The identity element is 0.

Each element $a \in Z$ has an inverse, which is $-a$, since: $a + (-a) = (-a) + a = 0$

Therefore, Z under addition forms a group because every element has an inverse.

Option C: A semigroup has closure, associative, and identity properties.

False: A semigroup doesn't need to have an identity property. It does not require an identifying component.

Example: The set of positive integers $Z^+ = \{1, 2, 3, \dots\}$ under multiplication forms a semigroup.

Closure: Multiplying two positive integers always results in another positive integer.

Associativity: For any $a, b, c \in Z^+$, we have:

$$(a * b) * c = a * (b * c)$$

However, there is no identity element in this semigroup because there is no number in Z^+ such that $a * e = e * a = a$ for all $a \in Z^+$.

Option D: A quasigroup has closure property.

True: A quasigroup follows the closure property—this means that any operation upon elements from the set outputs another element from the given set.

Example: The set $Z_5 = \{0, 1, 2, 3, 4\}$ under the operation of modular addition forms a quasigroup.

Closure: For any $a, b \in Z_5$, the result of $a + b \text{ mod } 5$ is still within Z_5 . For example: $3 + 4 \text{ mod } 5 = 2$. This set is a quasigroup because it satisfies closure.

Option A:

A, B, and D are correct.

Q83 Text Solution:

Issues:

Pointer Arithmetic:

$j = j * 2$: This line tries to multiply the pointer j by 2, which is illegal. In C, you cannot directly multiply or divide pointers by integers. Pointer arithmetic in C allows the addition or subtraction of integers to/from pointers, but not multiplication or division.



$k = k / 2$: Similarly, this line tries to divide the pointer k by 2, which is also illegal.

Conclusion:

This code will result in a compilation error due to the illegal pointer operations ($j = j * 2$; and $k = k / 2$);).

Correct Answer

Option D: Illegal use of pointer in main function.

Q84 Text Solution:

The given sets are:

$A = \{4n+2 \mid n \text{ is a natural number}\}$

$B = \{3n \mid n \text{ is a natural number}\}$

Find $A \cap B$, the intersection of sets A and B, and match it with the given options.

The general form for elements in A is $4n+2$, meaning the elements of A are numbers of the form 6, 10, 14, 18, ...

The general form for elements in B is $3n$, meaning the elements of B are numbers of the form 3, 6, 9, 12, 15, 18, ...

The intersection $A \cap B$ would be the common elements between these two sets. The first common number is 6, followed by 18, 30, etc., which form an arithmetic progression with a difference of 12.

So, $A \cap B$ contains elements of the form $12n-6$.

Option D is $\{12n-6 \mid n \text{ is a natural number}\}$, which matches our result.

Q85 Text Solution:

A. Simple recursive approach for calculating the Fibonacci number

This method causes the number of recursive calls to expand exponentially because it recalculates the Fibonacci numbers many times.

Time Complexity: $\Theta(\phi^n)$

Matches: IV

B. Memoization to compute Fibonacci number

Memorization reduces complexity by preventing unnecessary calculations and storing intermediate Fibonacci results.

Time Complexity: $O(n)$

Matches: III

C. Computing Fibonacci numbers using integer addition and multiplication:

Fast doubling or matrix exponentiation, which only requires integer operations, is used by efficient algorithms.

Time Complexity: $O(\log n)$

Matches: II

D. Optimal bitonic tour:

Dynamic programming solves the bitonic tour issue, which usually has a quadratic time complexity.

Time Complexity: $O(n^2)$

Matches: I

The correct option is B.

Q86 Text Solution:

(B) Both Statement I and Statement II are incorrect.

Statement I: If f and g are two functions and $f = O(g)$ but $g \neq o(f)$; , we say that the growth rate of g is smaller than that of f

This statement is incorrect. The notation $f = O(g)$ indicates that the growth of the function f is bounded above by the growth of g , up to a constant factor, for sufficiently large values of the input. The phrase $g \neq o(f)$ means that g does not grow strictly slower than f . Therefore, if $f = O(g)$ but $g \neq o(f)$, it means that the two functions grow at roughly the same rate, rather than one growing significantly faster or slower than the other. This contradicts the statement's claim that the growth rate of g is smaller than that of f .

Statement II: The class of all decision problems decided by a TM in exponential time, that is $O(2^k)$, k being a constant.

This statement is also incorrect. The class of decision problems solvable by a Turing Machine (TM) in exponential time is denoted by EXP. This class typically refers to problems solvable in time $O(2^{\text{poly}(n)})$, where "poly(n)" means some polynomial function of the input size n . The statement incorrectly suggests that exponential time corresponds to $O(2^k)$, where k is a constant. However, in computational complexity theory, exponential time complexity depends on the input size n , not a fixed constant k . Thus, the



exponent in exponential time is a polynomial function of n , making the given statement incorrect.

Q87 Text Solution:

Analysis of Problem -:

- 256 Mb DRAM is organized as $32M \times 8$ memory externally and as a $16K \times 16K$ square array internally.
- Each row must be refreshed at least once every 50 milliseconds.
- Refreshing one row takes 100 nanoseconds.
- We are asked to find the fraction of the total memory bandwidth lost to refresh cycles.

Solution: -

Calculate the Total Refresh Time for All Rows -

In this scenario, we have 16,000 rows (16K rows) to refresh, and each row takes 100 nanoseconds (ns) to refresh.

To find the total refresh time, we use the formula:

$$\text{Total Refresh Time} = \text{Number of Rows} \times \text{Time to Refresh One Row}$$

The number of rows =

$$16 \times 1024 = 16384 \text{ rows}$$

$$\text{Total Refresh Time} = 16384 \times 100\text{ns} = 1638400 \text{ ns}$$

Convert nanoseconds to milliseconds:

$$\text{Total Refresh Time} = 1638400\text{ns} \div 1000000 = 1.6384 \text{ ms}$$

Understand the Context of Refresh Cycles

Now, we assume that each refresh cycle takes 50 milliseconds (ms). This is the total time available in which the DRAM can be accessed.

Calculate the Fraction of Time Lost During Refresh

To find the fraction of time lost during the refresh cycles, we will use the formula:

$$\text{Fraction of Time Lost} = \frac{\text{Total Refresh Time}}{\text{Refresh Cycle Time}} \times 100\%$$

Now substituting the values:

Total Refresh Time: 1.6384 ms

Refresh Cycle Time: 50 ms

Substituting into the formula:

$$\text{Fraction of Time Lost} = \frac{1.6384}{50} \times 100\%$$

$$\text{Fraction of Time Lost} = 0.032768 \times 100\% \approx 3.2768\%$$

Rounding this value gives us approximately **3.3%** as the fraction of total memory bandwidth lost due to refresh cycles. Thus, during each refresh cycle of 50 ms, around 3.3% of the time is spent refreshing the rows, impacting the memory's available bandwidth.

Q88 Text Solution:

Both Statement I and Statement II are correct

Statement I: Subsystem models show logical grouping of objects into coherent subsystem. This statement is **correct**.

Subsystem Models

Subsystem models are a type of system representation that organize related objects into meaningful subsystems. These subsystems mirror the structure of the overall system and are frequently illustrated using class diagrams.

Coherent Subsystems

A coherent subsystem is a logical grouping of objects that are closely related and interact with one another. For example, in an e-commerce platform, all classes related to payment functionality can be grouped into a "Payment" subsystem.

Statement II: State machine models show how objects change their states in response to events. This statement is also **correct**.

State Machine Models

State machine models depict the transitions an object undergoes between different states in response to particular events. They provide a visual representation of how the behavior of an object changes due to external triggers.

Q89 Text Solution:

Both A and R are true and R is the correct explanation of A

An XOR linked list is a clever variation of a doubly linked list that uses a single pointer in each node to store the addresses of the previous and next nodes. The pointer in each node is an XOR of the addresses of the previous and next nodes.



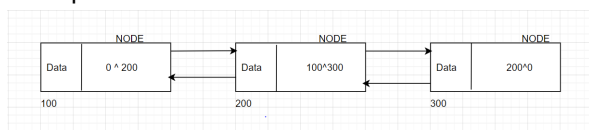
Concept:

In a traditional doubly linked list, each node stores two pointers:

1. A pointer to the previous node.
2. A pointer to the next node.

Instead of using two separate pointers in an XOR linked list, we store the XOR of the previous and next node addresses. This reduces the space required to store the list by half for pointers, as only one pointer is used instead of two.

Example: -



Let us take three nodes with addresses 100, 200, and 300.

every node stores the XOR of previous and next node addresses.

To get the next and previous nodes we have to use the formula -

Next node address = (address of previous node) ^ (current stored memory address)

Previous node address = (current stored memory address) ^ (address of next node)

Assume we are on a node with address 200 and now want to go to the next node apply the above formula -

Next node address = $100 \wedge 100 \wedge 300 = 0 \wedge 300 = 300$ (XOR operation: $1 \wedge 1 = 0$ and $0 \wedge 1 = 1$)

Hence we got the next node address 300 as depicted in the diagram.

Now if we want to go to the previous node from the 200 address node then apply the above formula -

Previous node address = $100 \wedge 300 \wedge 300 = 100 \wedge 0 = 100$ (XOR operation: $1 \wedge 1 = 0$ and $0 \wedge 1 = 1$)

Hence we got the previous node address 100 as depicted in the diagram.

So from the above example, we can conclude that although having one pointer, we can move in both directions using XOR operation.

Q90 Text Solution:

(A) Cross compiler

Cross Compiler -

A cross-compiler is a specialized compiler that produces executable code for a platform different from the one where it is being used. Essentially, it operates on one system (referred to as the host) but creates code that runs on another system (referred to as the target).

Example:

A cross-compiler running on a Windows computer can generate code for an ARM-based embedded device. The source code is translated into a binary format compatible with the ARM architecture, even though the compiler runs on a machine with an x86 architecture.

Q91. Text Solution:

To calculate the main memory hit ratio, we use the following formula:

$$\text{Hit Ratio} = \frac{\text{Number of Hits}}{\text{Total Memory Accesses}}$$

To calculate the total number of hits, we first need to determine how many memory accesses are made in total and then subtract the number of page faults from that.

Total Number of Memory Accesses:

The table T has 1024 columns and 17 rows, meaning the total number of memory accesses (since each element in the table must be accessed once) is:

$$1024 \text{ columns} \times 17 \text{ rows} = 17,408 \text{ memory accesses}$$

Number of Hits:

For each memory access, if there is no page fault, it's considered a hit.

Since there are 17 page faults, the total number of hits can be calculated as:

$$\text{Total Hits} = \text{Total Memory Accesses} - \text{Page Faults}$$

$$\text{Total Hits} = 17,408 - 17 = 17,391$$

$$\text{Hit Ratio} = \frac{17,391}{17,408} \approx 0.999$$

The main memory hit ratio is approximately 0.999, or 99.9%.

Q92. Text Solution:

Since T is stored in column-major format, the table will be accessed by columns, and each



column will be processed by adding up its elements.

The table has 17 rows and 1024 columns, and the main memory consists of 16 pages, each with a capacity of 1024 elements. This implies that a single memory page can hold up to 1024 elements from table T.

For each column:

There are 17 elements per column.

A memory page can store up to 1024 elements, so a single page can accommodate the full content of 60 columns (since $17 \times 60 = 1020$ elements) along with 4 additional elements from the 61st column (as $1020 + 4 = 1024$).

Page Fault Calculation:

When we access the first element of the first column, there will be an initial page fault because the data needs to be loaded into memory.

However, for subsequent accesses to the 2nd, 3rd, ..., and 17th elements of the first column, these elements will already be loaded into memory (as they are all stored in the same page). Thus, no additional page faults will occur for these elements.

In fact, 60 complete columns along with 4 elements from the 61st column will be loaded into the memory in the first page, meaning no page faults will occur for the first 60 columns. Once we access the 5th element of the 61st column, it won't be in memory, resulting in another page fault as this element will be loaded into memory along with the next batch of 1024 elements.

Conclusion:

For every 60 columns, there will be 1 page fault. Therefore, the total number of page faults is approximately:

$$\frac{1024}{60} \approx 17$$

Thus, the total number of page faults is 17.

Q93. Text Solution:

Let understand question with an example - Consider 3X3 matrix as shown in figure :

	c ₁	c ₂	c ₃
r ₁	10	40	70
r ₂	20	50	80
r ₃	30	60	90

Consider memory size 2 pages of 3 words i.e. a single page can store 3 words.

According to question we have to calculate sum of each column c₁, c₂ and c₃.

First we consider that table is stored in row major format. It means that table will be accessed by row and addition will be performed row by row. Elements will be accessed from main memory where LRU is used for page replacement as shown in figure :

frame 2		Access r2 for value 20	Access r2 for value 20	Access r1 for value 40	Access r1 for value 40	Access r3 for value 60	Access r3 for value 60	Access r2 for value 80	Access r2 for value 80
frame 1	Access r1 for value 10	Access r1 for value 10	Access r3 for value 30	Access r3 for value 30	Access r2 for value 50	Access r2 for value 50	Access r1 for value 70	Access r1 for value 70	Access r3 for value 90
	*	*	*	*	*	*	*	*	*

After applying LRU we can see that there are 3X3 = 9 page faults.

Every time one page will be missing and need to replace as per LRU policy. So all will be miss. Hence for $17 * 1024 = 17,408$ elements of T table and 16 pages of 1024 size memory, 17408 times there will be page access.

Hence there will be 17,408 page faults.

Q94. Text Solution:

Number of faults in column - major = 17

Number of faults in row - major = 17,408

Ratio row- major to column- major = $17408/17 = 1024:1$

Q95. Text Solution:

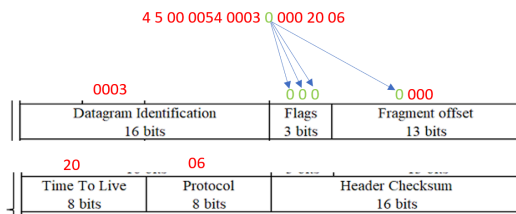
As for row major access every time one page will be missing and need to replace as per LRU policy. So all will be miss.

We can easily conclude that there will no (zero) main memory hits.

Q96. Text Solution:

← 4 5 00 32 bits 0054 →			
VER 4 bits	HEL 4 bits	Service Type 8 bits	Total Length 16 bits





45000054000300002006... this header information of IP datagram will be distributed as shown above.

It is in hexadecimal so every number represents 4 bits.

First hexadecimal digit 4 represents version of data packet. 4 in hexadecimal is actually represented as 0100 in binary.

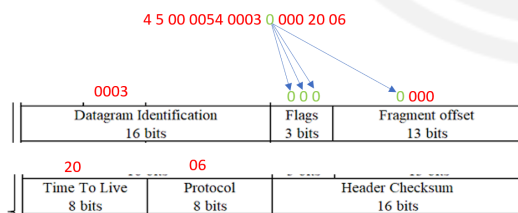
Since the first 4 bits of the datagram (0100) correspond to **IPv4**, this datagram is an **IPv4 packet**.

Now observe that flag requires only 3 bit and we know a number in header information represents 4 bits. Hence 0 (green colour) when represented in binary is 0000. So 3 bits are given to flags and other bit is added in fragment offset.

Now according to distribution **Time to Live field** has value 20 in hexadecimal i.e. **32 bytes** ($2 \times 16^1 + 0 \times 16^0$)

Hence, the packet can travel to 32 more routers.

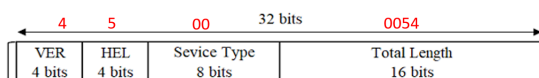
Q97. Text Solution:



From figure we see that protocol field has 06 information i.e. 6 value in decimal. ($0 \times 16^1 + 6 \times 16^0$)

So TCP protocol has 6 value for protocol field. Hence answer is **TCP protocol**.

Q98. Text Solution:



To the length of datagram is represented by 0054 as shown in diagram.

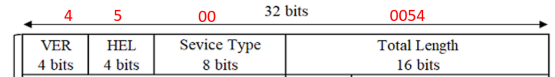
Now convert 0054 into decimal

$$(0054)_{16} =$$

$$0 \times 16^3 + 0 \times 16^2 + 5 \times 16^1 + 4 \times 16^0 = 84 \text{ bytes}$$

Hence size of datagram is 84 bytes.

Q99. Text Solution:

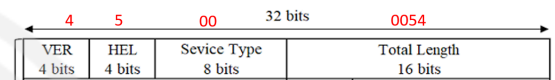


Head Length is 5, but it is in hexadecimal.

Convert it into decimal, so head length is $5 \times 16^0 = 5$ in decimal representation.

Now, Header length = $5 \times 4 = 20$ bytes

Q100. Text Solution:



$$\text{Total length} = (0054)_{16} =$$

$$0 \times 16^3 + 0 \times 16^2 + 5 \times 16^1 + 4 \times 16^0 = 84 \text{ bytes}$$

Now, Header length = $5 \times 4 = 20$ bytes

Hence Data carried by packet (Payload length) = $84 - 20 = 64$ bytes.

Efficiency = payload length/ total length = $64/84 = 0.7619$

or % Efficiency = **76.19%**





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